A Complete, Certified Decode of a Production Language Model: Every Dimension of GPT-2's State Space Priced, Named or Proven Word-Less, and Spoken Back

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v1.1, 2026-07-08; revised per external agentic review (paperreview.ai) of the 2026-07-06 v1 — adds multiplicity / false-discovery control, floor-construction sensitivity, transplant boundary×regime generality, door/core rotation robustness, seam perturbations, a student-methods appendix, a no-gloss scope reframe, a glossary, and expanded related work; changes summarized in REVISION_NOTE.md and answered point-by-point in REVIEWER_RESPONSE.md. No headline number changed — every revision scopes a claim, none overturns one. All frozen, hash-stamped artifacts are public: concept DOI [10.5281/zenodo.21230107](https://doi.org/10.5281/zenodo.21230107) (always resolves to the latest version; this v1.1 receives its own versioned DOI), code github.com/wpferrell/babel-codec-gpt2, mirror huggingface.co/wpferrell/babel-codec-gpt2.

Abstract

We present the first complete, certified, bidirectional decode of an entire production language model: every dimension of GPT-2's residual state space priced, named or proven word-less, and spoken back. The grain is stated up front: the residual state at each of GPT-2's 13 layer boundaries — not the within-block sublayer mechanisms (attention/MLP interiors enter only as attribution evidence), so "complete" means complete *at boundary grain against these floors*, the program's pre-written success criterion. On GPT-2 (124M, fp32), across all 13 residual-stream boundaries in three text regimes (prose, code, repetition) — a 39-cell battery — a decoder built from the model's own certified channels replaces the full hidden state at every cell with behavior held at or under the model's own per-regime noise floor: 39/39 cells on the primary (norm-relative) meter — ratified mid-program under three pre-registered sanity gates (§2) — 36/39 on the stricter legacy meter (the 36/39 is robust to floor construction; the last three cells close only under the *full* norm-relative recalibration — at any partial norm-scaling they reopen — sensitivity table in Appendix C), with zero unexplained nats remaining on the primary meter (down from 11.2 at the first verdict). Success is judged by *substitution* — forward KL against pre-registered floors — never by narrative, and the account survived six consecutive pre-registered NOT-YET verdicts before the band was met. On top of the account sits a certified bidirectional codec for GPT-2's residual-stream language (the BABEL codec): every one of its 351 channels is adjudicated against sigma-matched nulls — 53.6% carry an explicit English gloss and 46.4% are *certified word-less* (a tested outcome, not a gap; the 53.6% is the frozen per-gate rate under the L1 σ-matched-null battery at N = 20 draws — under explicit multiplicity control the named fraction is materially lower and correction-dependent: ≈ 26% at channel-level Benjamini–Hochberg $q = 0.05$, ≈ 7% under a strict per-gate BH, Holm FWER ≈ 1 channel, with a ≈ 9% strong core that no correction removes and a ≥2-of-3-regime rule that already bounds the false-named fraction near 10% — Appendix B); the layer-to-layer composition of the named fields is linear-certified at all 36 seam-cells (12 seams × 3 regimes) — Stage-2 perturbations (§6.2) show this survives field-basis rotation and $\pm\varepsilon$ dictionary jitter, and independent per-seam field re-derivation at all 33 propagation seams; the lone exception is the first (embed→L0) *rewrite* seam, whose linearity rests on the certified global fields; an exact algebraic inverse (gloss → state) is well-posed at 39/39 cells; and the language runs both ways behaviorally: read-then-re-encode preserves behavior at 39/39 cells, transplanting a read-out gloss between contexts carries 94.7% of the next-token behavioral gap (matched-random: 18.6%; 16 prose pairs at one mid-stack boundary; a Stage-2 boundary×regime table, §6.4, confirms this is not boundary-special — median closure 0.94–0.98 in prose and 0.82–0.98 in repetition across early/mid/late boundaries and 0.89 in code at late depth, though code is heavy-tailed at early/mid depth) — with the residual 5.3% *certified* to lie in un-adjudicated dark mass outside the entire certified channel set: adding the certified door channels to the transplant closes exactly 0.0 more (§6.6), and *characterized* (§6.7): diffuse across the 329-dim dark complement (no subspace of rank ≤ 256 closes 80% of the gap; a random 256-dim dark slice nearly matches the best) and mostly word-less (6 of its 8 largest carriers certified no-gloss, the other 2 faintly nameable only through non-vocabulary channels) — and hand-editing single English fields steers the model in that field's own vocabulary for 3 of 4 named axes (the fourth, the executable "onset rung", is certified *unusable as a steering lever at both tested doses*: its tiny onset effect — $|A| \approx 0.004$ at $\pm 3\sigma$, ≈ 0.010 at $\pm 6\sigma$ — clears no internal readout, does not separate from an honest $N = 20$ matched-random floor at either dose (at $\pm 3\sigma$ the effect sits inside the floor's own draw-to-draw spread across two pre-registered $N = 20$ draws, §6.6), and scales sub-linearly, so pushing harder buys nothing; one candidate certified listener, 2.8% over a 12-draw multiplicity null, reads its output downstream, §6.6–6.7). The hardest object — a layer-5 repetition-maintenance computation that defeated every read and every circuit splice — is closed by a 1.18M-parameter *linear* surrogate that certifies on never-seen repeat periods while higher-capacity students fail the memorization falsifier. No prior or concurrent work — the nearest being Anthropic's Natural Language Autoencoders and the Cycle-Consistent Activation Oracles line — provides all four of the claim's defining properties: whole-model coverage with a priced remainder, behavioral certification against matched nulls, a constructive route through the model's own certified channels, and a bidirectional behavioral round trip (§7, Table 1, adjudicates this row by row and commits to amending the claim should a counterexample surface). All results were produced and are reproducible on one workstation GPU; every claim carries an evidence hash.

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1. Introduction

1.1 The question

Can the internal state of a production language model be *fully* accounted for — not sampled, not illustrated by cherry-picked features, but priced dimension by dimension at every depth, against a bar the model itself sets — and can the accounted-for part then be translated into English and back?

This paper answers that question for GPT-2 124M at a defined grain: the residual stream ("bus") at its 13 layer boundaries, in three text regimes. The answer is yes, with every caveat stated and priced: yes at 39/39 boundary–regime cells on the primary meter (36/39 on the stricter legacy meter, the 3 residual gaps being priced meter geometry, not model behavior); yes for translation under a definition in which *"this channel provably carries no word"* counts as a completed, tested translation outcome (46.4% of channels); and yes behaviorally, including a human-edit tier in which turning up a named "naval/warship" field makes the model predict *amphib, sunk, ashore, reefs, sailed, submarine*.

1.2 What is new

Reading activations with language is no longer new: a lineage runs from logit-lens-style projections through trained translators (LatentQA, ParaScopes, DecoderLens, Patchscopes), and, in concurrent work published this spring, Anthropic's Natural Language Autoencoders (NLA) and the Cycle-Consistent Activation Oracles (CCAO) train verbalizer/reconstructor pairs that explain single activations (§7). What has not existed is an *account*: a claim of the form "this is everything, at this grain, and here is the bill for what resisted." Stated in its locked form, the claim of this paper is that it is **the first complete, certified, bidirectional decode of an entire production language model**. Four properties define that claim — whole-model coverage with a priced remainder, behavioral certification, a constructive route through the model's own certified channels, and a bidirectional behavioral round trip — and they are exactly the first four contributions below. §7 (Table 1) adjudicates every prior and concurrent line against all four and states the standing commitment to amend the claim if any work is found to provide them. The contributions:

1. **Completeness with a price.** Every one of 39 (boundary × regime) cells is priced by full-state substitution against a pre-registered per-cell floor. The completeness score and the unexplained mass (in nats) are published per cell, on two meters, with the meter change itself pre-registered, sanity-gated, and reported on both sides (Fig. 1, Fig. 2).
2. **Certification rather than plausibility.** Every claim is a mechanical verdict against bands locked before measurement. Explanations are never scored by whether they sound right; channels that fail naming are published as CERTIFIED-NO-GLOSS — a first-class result with, to our knowledge, no published analogue.
3. **A constructive route.** The decoder is assembled from subspaces certified as *necessary* on the model's own computation (doors, core fields, folded reads, executable rungs) —

not from a trained external translator. Two of the headline results — the internal-alphabet result (§4.4) and the 36/36 seam-linearity law (§6.2) — are therefore claims about the *model*, not about a translator's competence.

4. **A behavioral round trip.** Translation quality is scored by what the model *does* after re-encoding — substitution KL, cross-context meaning transplant, and human edits of single English fields — with pre-registered falsifiers and matched-random nulls at every tier (§6.4).
5. **Negative results as method.** The account was reached through six consecutive pre-registered NOT-YET verdicts (§1.3), three certified impossibility results on one object (§5), a capacity-hurts falsifier that caught its own would-be false positives (§5.4), and one instrument bug caught by its own replay gate (§3.6). We argue this discipline is the paper's most transferable contribution.

1.3 The six NOT-YETs, told as the method they are

The completeness hypothesis (H-OPEN6-a) was written once, with a band requiring all 39 cells at or under floor, and an escalation rule binding the program to publish the gap table and stop — relaxing nothing — whenever the band was missed. It was missed six times: at the campaign verdict (19 gap cells, 11.2 excess nats), and at decoder rounds V2 (9.8), V3 (7.5), V4 (7.5), V5 (6.8), and V6 (3.1). Each NOT-YET is a dated, hash-stamped record of exactly what remained dark and what it cost; each round's attack was chosen from the previous round's gap table; and the verdict text never changed. When the band was finally met at V7 — 39/39, zero unexplained nats, as the pre-registered 9% *underdog* outcome — the result inherited the credibility of the six honest failures that preceded it. Figure 1 is this history. We commend the pattern: completeness claims are cheap; completeness claims with six public prior refusals are not.

1.4 Scope, stated plainly

One model (GPT-2 124M), one grain (boundary-level residual state; within-block mechanisms enter only as attribution evidence), three regimes, floors defined by the model's own output sensitivity to state noise. "Complete" always means *complete at this grain against these floors* — the defined success criterion of the program, written before the data. §8 prices every scope edge.

1.5 Glossary (one page; added v1.1 per review — the prose is dense with coined terms)

term	one-line definition
boundary / bus	The residual stream `h_b(t)` read at a layer boundary `b ∈ {0..12}` (BUS[0] = embedding output, BUS[12] = final pre-unembedding).

term	one-line definition
	The grain at which the account is "complete."
cell	One (boundary × regime) pair; the account has 39 = 13 boundaries × 3 regimes.
meter / substitution	The scoring rule: rebuild $\hat{h}_b$, substitute it for h_b, score forward KL(clean ‖ substituted) in nats. Behavior, not narrative, is the judge.
floor	The pass bar for a cell = the model's own output KL sensitivity to matched state noise. *Legacy* floor: prose-anchored (0.1871 nats). *Recalibrated / primary* floor: the same noise re-measured at the per-cell state norm. A cell PASSES iff substitution KL ≤ floor.
door (Q_union)	The minimal-rank subspace through which a module group must write to preserve behavior (≈ 385 of 768 dims); certified by restricting writes to a rank-k basis and demanding KL ≤ floor.
core (C)	The 19-dim shared ensemble core of the certified doors; individuated, semantic, load-bearing.
front door (m0)	MLP0's certified minimal write into MLP1 — the detokenization bottleneck.
corridor (Q35)	A boundary-persistent 35-dim dark object recurring across the biggest dark rooms; "located first, named later."
fold / folded read (O_r48)	A rank-48 orthonormal read of the residual complement at a cell, within the ≤ 64-unnamed-dims allowance.
rung (executable)	Where no static read suffices (rep b5/b6/b7), a trained ~1.18M-param *linear* map that *computes* the missing object from readable inputs; certified by substitution on never-seen repeat periods.
dark mass / dark room	Residual variance not read by any certified channel; peaks early-mid, most enumerable there.
the wall	The BUS[5] repetition-maintenance computation that resisted every read and splice; closed only by running a rung.
gloss	An English word/label a channel earns by clearing the naming battery (CH-WU / CH-INT / CH-FIELD) against σ-matched nulls in ≥ 2 of 3 regimes.
certified no-gloss (word-less)	A channel that fails the naming battery against its nulls — a *tested* outcome ("this glyph is decoration"), scoped to the battery at its budget (see §8.4).
seam / grammar	The layer-to-layer map $\varphi_b \to \varphi_{b+1}$ of the named fields; certified LINEAR at all 36 seam-cells.
transplant	Encoding context A's readable gloss into context B and measuring how much of the A↔B behavioral gap it carries (94.7% vs 18.6% matched-random).
off-span / manifold-bound	Behavior when a gloss is pushed beyond the visited manifold; "manifold-bound" = indistinguishable from matched noise off-span.
gauge, not a lever	A channel whose value *reads out* a quantity but cannot *steer* it (the onset rung).
N_open / N_grain	Cells passing with zero unnamed dims (`N_open`) vs with ≤ 64 located-but-unnamed dims (`N_grain`, the program's success criterion).
birth cells	The six embedding-era cells (BUS[0..1] × 3 regimes), closed by doors + the token-surface term.
twin (shuffled-target)	A control student trained on permuted targets; a real student must beat its twin by a preset factor.

term	one-line definition
byte-replay gate	A reused component must reproduce its banked anchors to the digit before its outputs are trusted.
DEAF / glitch axis (b2_d0)	The canonical inert channel: its response is dwarfed by its own nulls (field 0.87 vs null95 ≈ 118).
decoupling law	Variance ≠ behavior in both directions (high-variance / low-behavior outliers; low-variance / high-behavior wall).

2. Definitions: the state, the meter, the verdicts

State. `h_b(t)` is the 768-dim residual state at boundary `b ∈ {0..12}` (BUS[0] = embedding output; BUS[12] = final pre-unembedding state), position `t`. All runs: fp32, eager attention, TF32 off, fixed seeds.

Regimes. prose (WikiText, fresh token windows disjoint from all fit data), code (HumanEval concatenation, fresh windows), repetition (seeded synthetic induction streams, period-64 repeated segments; held-out seeds and periods reserved for falsifiers). Holdout per regime: 16×512 tokens.

The meter (substitution). At cell `(b, regime)`: capture clean logits on the holdout; rebuild `ĥ_b` from decoder reads only; substitute `ĥ_b` for `h_b` in the forward pass; score forward KL (clean ‖ substituted), in nats, averaged over positions (repetition rungs metered on the repetition-era positions, per the certified per-cell convention). Identity substitution must be *exact zero* (KL ≤ 1e-9, max |Δlogit| ≤ 1e-4) at every cell before any verdict is read — the plumbing gate.

Floors (two meters, both permanent). A cell PASSES iff substitution KL ≤ its floor. *Legacy:* per-(boundary, regime) floors from matched noise injections (the J1 bank), anchored to the prose cross-seed ensemble floor `EPS_KL = 0.1871` nats (T6). *Recalibrated (primary):* norm-relative floors — the same noise construction scaled by the measured per-cell state norm (`_v5_floors_recal.json`, hash 71549ae3...) — adopted mid-program under three pre-registered sanity gates: all 26 legacy floors replicated to the digit first; the early-repetition floors (proven genuine model physics, §5.1) were not allowed to relax; the wall cell was not allowed to pass by bookkeeping. Both meters are reported everywhere; cells that pass only under recalibration are labeled *meter corrections, not model discoveries*.

The two completeness rungs. `N_open`: cells passing with *zero* unnamed dimensions (named fields + certified doors + token-surface term only). `N_grain`: cells passing with at most 64 unnamed-but-located dimensions per boundary (the corridor and folded reads; the allowance arithmetic is shown per cell). `N_grain = 39/39` is the program's pre-written success criterion.

Verdict discipline. Every experiment is one pre-registration block (gap-scan, design, exact bands, stated bet with odds, failure branches, budgets) locked in an append-only pen before any measurement, and one results block with mechanical verdicts. Bands may be sharpened, never weakened; missed favorites are logged loudly (the record includes rounds where 0/5 favorites hit, and rounds where 7/7 did).

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3. The instrument: how the decoder was built and policed

3.1 The certified alphabet (what "the model's own channels" means)

The decoder's reads are subspaces certified as necessary at the model's noise floor, built up over the program's instrument series (all pre-registered; hashes in Appendix A):

- **Doors (Q_union, 385 dims ≈ 50% of d=768).** The minimal-rank subspaces through which module groups must write to preserve behavior: joint attention-writer door k*≈128, MLP-interior k*≈240, union ≈385 — measured by restricting group writes to a rank-k orthonormal basis and demanding KL ≤ floor. Per-module necessity is ~1 dim; load pools at the group level.
- **The core (C, 19 dims).** The shared ensemble core of certified doors across cert sets: individuated (19/19 distinct), semantic (17/19 carry content classes), analog (0/19 discrete), load-bearing (excising the 19 dims hurts more than ablating all 12 attention writes at once).
- **The front door (m0, k*=13 prose / k*=1 repetition).** MLP0's certified minimal write into MLP1 — the detokenization bottleneck; its single repetition-era coefficient is one of the three inputs of the executable rungs (§5.4).
- **The corridor (Q35, 35 dims).** A boundary-persistent dark object discovered as ~35 distinct directions recurring across the five biggest dark rooms (cross-boundary direction matches to 0.9999). Located first, named later: 26/35 now carry glosses (§6.1).
- **The token-surface term (wte).** A ridge read from the current token's embedding to the content-complement residual — the only *fitted* read in the entire named decoder, frozen with its hash. It completes the birth boundaries (§4.3) and provably does not help late (a pre-registered null: surface ≠ late content).
- **Folded reads (O_r48, per-cell).** Rank-48 orthonormal reads of the residual complement at late/code cells, inside the ≤64 unnamed-dims allowance (arithmetic shown per cell, 61–62 ≤ 64).
- **Executable rungs (S9x, 3 cells).** Where no static read suffices (rep b5/b6/b7), a printable ~1.18M-parameter *linear* map computes the missing object from readable inputs {layer-2 state, current token, m0's certified coefficient} and the decoder runs it (§5.4). The rungs are the only trained modules in the decoder, and they are certified by substitution on never-seen repeat periods with a memorization falsifier.

3.2 The naming battery (how a channel earns a gloss, or is proven word-less)

Each channel (corridor direction, core field, folded dim at ≥1% share) is snapped causally (±1σ/±2σ at its top-traffic positions) and judged on three response channels: its own unembedding-vocabulary image (CH-WU), logit-lens contrasts at downstream boundaries (CH-INT), and the antisymmetry of its downstream field response (CH-FIELD). *Every* channel is judged against sigma-matched random nulls — random directions snapped at the same magnitude — and a gloss requires stability in ≥2 of 3 regimes (regime-specific naming allowed for folds, labeled). Channels that fail *against the null* are CERTIFIED-NO-GLOSS: random directions of the same loudness move the model *more*. The glitch axis b2_d0 is the canonical example: response dwarfed by its own nulls on every channel (e.g., field response 0.87 vs null95 ≈ 118) — inert, not weak. Below-1% folded dims are certified collectively as noise-floor residue (per-cell share 0.065–0.191).

3.3 Null design (the three families)

Sigma-matched nulls for naming (above); *matched-random substitutions/edits* for behavioral tiers (a random read/edit of the same per-position norm, the discriminator for §6.4); *shuffled-target twins* for trained modules (an identical student trained on permuted targets — any student that cannot beat its twin by the pre-set factor is void). Where a control itself carries a lesson (the operator axis, §6.4), the lost sub-prediction is logged as a finding.

3.4 The falsifier culture

Held-out repeat *periods* (not just positions) so a student that memorizes trajectories is caught; the capacity-hurts falsifier (higher-capacity students must not be allowed to win by memorization — and in fact they lose, §5.4); pre-registered failure branches that fire as contingencies, not weakenings; post-hoc analyses labeled post-hoc, opening questions rather than closing them.

3.5 Byte-replay gates

Every reused component must reproduce its banked anchors to the digit before its outputs are trusted (12+ replay gates in a typical round; "0 instrument discrepancies" is a logged, checked condition, not a phrase). The one instrument bug of the program's decoder week — vocabulary-channel nulls snapped at the wrong sigma — was caught by its own replay gate, diagnosed against the wrong hypothesis first (that too is logged), and re-run bounded with all 70 prior null quantiles replayed to ≤ 5e-5.

3.6 Reproducibility

Single GPU (NVIDIA RTX A4500, 20 GB): the seven decoder rounds cost 183.6 s – 4891.6 s GPU each; the four BABEL-codec stages 74.9 s – ~4600 s. Deterministic streams (exact token windows + seeds); atomic checkpoints; every frozen artifact hash-stamped (Appendix A); figures regenerate from the frozen JSONs alone (paper_figs/make_paper_figs.py).

4. Results I — the account: 39/39 cells at or under the model's own floor

4.1 The verdict

Fig. 1 (nat collapse) and **Fig. 2** (final heatmaps, both meters). At the V7 grain:

meter	N_open	N_grain	gap cells	unexplained nats
recalibrated (primary)	6/39	**39/39**	**0**	**0.000**
legacy (permanent)	6/39	36/39	3	0.283

The three legacy-only gaps (rep_b11 1.47×, rep_b12 2.57×, code_b12 1.93×) all close under recalibration and were pre-priced as meter geometry (the late repetition/code state norms grow; the legacy bar there reflects calibration geometry, not model sensitivity — §5.1). The six birth cells (BUS[0..1] × 3 regimes) pass at the *strict* zero-unnamed-dims rung: the certified door alphabet plus the token-surface channel fully account for the embedding-era state (repetition BUS[1]: doors alone 3.45 nats → +wte 0.061 vs floor 0.135). Ordinary prose is grain-open at every boundary; so is everything else — that is the claim.

Floor-sensitivity (v1.1, Appendix C; answering review Q1). The 36/39 legacy closure is robust to floor construction — those 36 cells pass under essentially any reasonable floor. The step from 36/39 to 39/39 is entirely a property of the adopted norm-relative meter: sweeping a fractional norm-scaling floor `floor_β = floor_legacy·(floor_recal/floor_legacy)^β` ($\beta = 0$ legacy, $\beta = 1$ the adopted recal) gives 36/39 for $\beta \le 0.5$, 37/39 at $\beta = 0.75$, and 39/39 *only* at the full $\beta = 1$. The three recal-only cells close because the late-layer state norm grows and the norm-scaled floor there is 1.5–2.7× the prose-anchored legacy floor (the ratio tracks ρ^2, ρ = regime/prose RMS) — a documented norm-geometry effect (§5.1), not new model behavior, but a genuinely meter-dependent closure that we label as such. Under the recal meter closure is 39/39 for any bar *loosening* ($c \ge 1$) but 34/39 at a 10 % tightening ($c = 0.9$); of the 39 cells, 14 pass comfortably (KL/floor ≤ 0.5), 12 moderately (0.5–0.8) and 13 tightly (0.8–1.0). Honest headline: **36/39 floor-construction-robust; 39/39 under the pre-registered norm-relative meter at full norm-scaling.** A bootstrap CI over holdout tokens is deferred to Stage 2 — only per-cell mean substitution KL is frozen (per-token KL was not stored), so token-level CIs require a GPU re-run (STAGE2_PROPOSAL.md).

The grain at which each cell closes is part of the account (Fig. 2 annotations; Appendix A): 17 cells close on named content alone (B2 + corridor + wte), 19 on folded r48/O20 reads within the allowance, and 3 — the repetition wall and its onset — only by *running* a certified linear rung.

4.2 The decoupling law: variance ≠ behavior, in both directions

The program's most load-bearing negative discovery, established across three rounds and then sharpened into a single-coordinate exhibit and a 1-vs-128 asymmetry:

- The five late "massive-activation" outlier directions hold 38–64% of late-tail variance and almost none of the behavior: an oracle read of the model's true coefficients along them moved the needle by only −0.13 median nats and crossed zero floors (V2); at one cell the oracle read made behavior *worse*, later diagnosed to a single behaviorally anti-correlated coordinate — banked outlier dim 1 at b11 — whose removal flips the read from harm to help (V3).
- The layer-5 repetition object is the mirror image: reconstructed at field-R^2 0.95, it still sat at 28× its floor (V2) — low residual variance, dominant behavioral mass.
- The asymmetry is quantified from both sides of one write: MLP0's repetition-era output is certified *one* dimension wide (k*=1, margin 63.5% after recalibration), while the object that write feeds carries behavioral content across ~128 dimensions (k_obj* ≈ 128, V5). One-dimensional to certify; hundred-dimensional to be.

Consequence: any decoder that ranks reads by variance mis-allocates on both ends. Ours did, until the meter said so.

4.3 The coordinate inversion: the dark mass speaks the token language

At the campaign's start the unread ("dark") mass was mapped: it peaks *early-mid* (BUS[2] 4.84 nats, BUS[2..6] 2.9–4.8), not at the output, and is most enumerable exactly there (k90 = 128 dims), while the late stack is bright with a small diffuse remainder. The five biggest dark rooms then failed every certified door read (medians 0.23–0.41 vs a 0.50 bar) but were *out-read by the raw current-token embedding* (0.42–0.76): the dark mass is surface-coded, current-token-conditioned content, ~85% MLP-written — which resolved, in place, why door-span coordinates cannot read it and why interior MLPs carry 53% of write variance while adding ~0 single-hop readability. The wte read term built from this finding closes the birth boundaries completely and cuts the rooms 8–50% — and adds ~nothing at BUS[10..12] (its pre-registered scope null: surface content is not late content; it even lowers late field-R^2 in repetition, 0.563 → 0.513 at rep_b12).

4.4 The internal vocabulary (BROAD-SURVIVES)

At the output vocabulary the 35-direction corridor is nearly mute (2/35 name via W_U). One layer downstream, the model demonstrably *computes* off it: the passthrough-excluded field probe showed sign-antisymmetric, null-beating door responses on 33/35 directions (V3); re-run with sigma-matched nulls — the magnitude control that deleted 14 of those 33 — the result survives at 19/35 on the strict field channel alone, 21/35 overall (V4C). Namability correlates with neither variance (r ≈ 0.11) nor behavioral coupling (r ≈ 0.19). There is a vocabulary in there that the output never voices; it became the target list for the naming campaign of §6.

4.5 The decoder predicts in flight

Frozen before generation, DECODER_V1 watched the model write and called: (P1) the regime of not-yet-generated text, 3/3; (P2) depth-4 field trajectories at median R^2 0.5016 (the honestly bet PARTIAL); (P3) a steering counterfactual — pushing the harm/casualty field pulls harm vocabulary into the output at +0.2168 (3.4 SE), within 2.5% of the banked prose anchor 0.22181, on self-generated text (baseline determinism 0.0). The narrated-generation transcript is an artifact (NARRATED_GENERATION_V1.md).

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5. Results II — the wall and its student

One object consumed five rounds and became the program's centerpiece: at BUS[5] in repetition, a 3.34-excess-nat cell (28× floor) that at campaign close held 30% of everything unread — and, after V5, 79% of it.

5.1 What it is

Repetition-specific (5.6× the prose variance), diffuse (159 of 364 complement dims for 90% variance), invisible to the current token (R^2 −0.22), to phase (0.02) and to position (0.04), but readable from the layer-2 state at R^2 0.55 — a computation carried forward, not a fact read off the input. It is causally load-bearing: ablating it doubles the probability of exiting the repeat, 15× the energy-matched random null. Its circuit is attributed: MLP0 — the certified detokenization front door — anchors 77.6% of its energy (5.3× a matched random basis; removing that one contribution drives exit probability 0.045 → ~1.0), supported by two lag-64 same-token heads (a0.h1, a0.h5) and three pure repeat-structure heads (a1.h7, a1.h4, a2.h10; path-patching shows their contribution survives swapping *which* segment repeats and dies when repetition is removed). The textbook induction heads sit *downstream*: they read this object, they do not write it. Separately, the early-repetition *floors* at b5–b7 were proven genuine physics — they survive norm-matching (2.16 nats) — while the impossibly strict *late* repetition floors were shown to be calibration geometry (~0.19 ≈ prose) and honestly re-priced (§2).

5.2 Three certified impossibilities

1. **It cannot be read** (V2/V3): every pre-registered read — current token, phase, position, earlier layer, corridor, upcoming-copy window, full 64-token period, a combined 13k-feature linear read, a bilinear grain — fails to beat the modest banked earlier-layer read (0.5525, replayed to the digit). Null certified.
2. **It cannot be spliced** (V4/V5): substituting the object with what its own certified writing circuit writes, pen-width enforced (k*=1), scores 4.256 nats — *worse than not reading it at all* (3.460) — and is beaten by 15/20 energy-matched random component

sets. The corrected rebuild under the recalibrated meter, with rank-matched nulls per variant, the energy books opened, and the skipped variant run, reproduced it byte-identically (4.25577). At full width the named circuit is component-specific (beats rank-matched nulls on both legs) yet still leaves 18–20× floor. Output-narrow is not object-narrow.

3. **Bigger students memorize** (V6): see below — the falsifier result.

5.3 The re-pricing that didn't move the wall

The meter recalibration (V5) moved the wall's own bars by <9% (rep b5 floor 0.1223 → 0.1279): early repetition fragility is model physics, and the pre-registered sanity gate that would have caught a wall-passes-by-bookkeeping outcome had nothing to catch. Three cells were honestly re-priced as geometry (labeled as such in every table); the wall stood.

5.4 The student (FQ-16): the wall's computation is linear

Fig. 4. Three students were trained by MSE to predict the object from readable inputs only {layer-2 state, current token, m0's k*=1 coefficient}, then substituted at BUS[5] and metered on *never-seen repeat periods* (SACRED): a linear map (1.18M params), an MLP (1.77M), an attention student (1.78M). The two high-capacity students fit training content nearly perfectly (train-R^2 0.98/0.97, within-seen KL ≈ 0.001) and *fail on new periods* (SACRED KL ≈ 0.35). The linear student — too small to memorize — generalizes: SACRED KL 0.11172 ≤ the 0.1279 floor, SACRED-R^2 0.77, 13.7× under its shuffled-target twin (1.536), with the repetition loop behaviorally intact (copy fidelity 0.9636). The certificate's thin margin was then discharged the pre-registered way: 5 fresh batches, 80 never-seen blocks — median 0.11814 (7.6% under floor), 4/5 batches under, worst 1.05× (band allowed 2×). Permanent label: *certified, re-confirmed, margin thin-but-stable*.

The onset cells (rep b6/b7 — 67% of the remaining dark after V6, and the cells we expected to name-and-exclude as raw physics) certified the same way, with wide margins and under *both* meters: SACRED 0.02644 ≤ 0.08057 (b6, 9.6× under twin, behavior 0.9995) and 0.06665 ≤ 0.10724 (b7, 4.9×, 0.9711). The "interactional physics" hypothesis for the onset is refuted at this grain; no exclusion convention was needed. A final lost bet, prized: dropping the m0 coefficient and retraining *improves* b5 slightly (0.11105) — {layer-2 state + current token} is the load-bearing alphabet, and the record says so in a re-scoped ledger row rather than a quiet edit.

Fragile floor, learnable object: both are measured facts, and they do not conflict. What five rounds proved impossible to read, remember, or forge is a *printable linear formula* in the right coordinates — found only because the falsifier was built to catch every easier illusion first.

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6. Results III — the BABEL codec: the dictionary, the grammar, the inverse, and the speak test

The BABEL codec was built and scored as four pre-registered cells; the definition (BABEL-COMPLETE) was fixed before any stage ran.

6.1 The dictionary (LEXICON_V3): 351/351 channels adjudicated

channel class	adjudicated	NAMED	CERTIFIED-NO-GLOSS
corridor directions	35/35	26 (74.3%)	9 (incl. the DEAF glitch axis b2_d0)
core fields	19/19	19 (100%)	0
folded dims (≥1% share)	297/297	143 (48.1%; incl. 15 regime-specific)	154
total	**351**	**188 (53.6%)**	**163 (46.4%)**

Below-1% folded dims are certified collectively as noise-floor residue per cell. The regime split beneath the fold fraction is itself a finding: repetition cells name richly (e.g., rep_b8 15/22, rep_b10 22/32) while code cells name sparsely (code_b9 2/9) — the folded geometry carries dense, regime-conditioned vocabulary in repetition and a thin, mostly noise-floor remainder in code. Every entry carries its per-entry evidence hash; every verdict is against sigma-matched nulls.

Multiplicity control (v1.1; answering review Q2). The counts above are the *frozen per-gate* adjudications: a channel earns a gloss when the same channel clears a σ-matched-null gate (obs > the 95th percentile of a 20-draw null, plus a ±1σ/±2σ dose-consistency filter) in ≥ 2 of 3 regimes. Because the paper reports 351 channels × 3 tests × 3 regimes, we add explicit false discovery / familywise control (Appendix B; harnesses `_rev_fdr.py`, `_rev_fdr_channel.py`, pre-registered before computing). Three complementary readings, on the 312 channels whose full per-gate statistics are frozen (297 folds — the dominant multiplicity exposure — + 15 corridor directions; frozen NAMED = 149/312 = 47.8 %):

control	q = 0.01	q = 0.05	q = 0.10
per-gate Benjamini–Hochberg (m = 2592 gates; strict)	1.9 %	**6.7 %**	16.3 %
channel-level BH (m = 312, conjunction-aware)	9.0 %	**25.6 %**	35.6 %
Holm–Bonferroni FWER (α = 0.05, per-gate)	—	0.3 % (1 channel)	—
global-null bound (assumption-free): expected false-NAMED ≈ 15.5/149 → FDP ≈ **10 %**			

The two BH rows bracket the truth and the gap is interpretable: per-gate BH treats a channel's nine gates as nine independent hypotheses and *discards* the ≥ 2-of-3-regime conjunction the rule requires, so it is a lower bound; the channel-level combination credits that conjunction. Most frozen names rest on gate clears that are individually marginal (obs just above the 0.05 σ-null gate) and are made credible by cross-regime replication rather than single-gate strength — so a strong ≈ 9 % *core* (q = 0.01) survives any correction, roughly half the frozen names survive channel-level BH at q = 0.05, and the built-in replication rule already holds the false-named fraction near 10 %. **The N = 20 frozen null is the binding limit:** the assumption-free empirical p is floored at ≈ 1/21 = 0.048, so continuous p below ≈ 0.05 are half-normal tail-model extrapolations from a single quantile; a *definitive* FDR needs a Stage-2 high-N (≥ 1000) σ-null re-draw (STAGE2_PROPOSAL.md). We retain the 53.6 %/46.4 % figures as the frozen gate-level record and re-scope every naming claim to "named / word-less under the L1 σ-matched-null battery at N = 20"; the 19 core fields (all named, individually adjudicated in an earlier round, T15) and ≈ 20 further corridor directions (V3/V4C) are outside this file's per-gate statistics, so the corrected overall 351-fraction is ≈ 23 % (out-of-file names uncredited) to ≈ 34 % (credited) at channel-level BH q = 0.05, versus the frozen 53.6 %.

Basis and rotation robustness (v1.1; answering review Q3). The necessity certificates behind the account are rank statements — inject a rank-k orthonormal *write* of an object's subspace and demand KL ≤ floor — so they are invariant to the orthonormal basis of that subspace *by construction* (the write is a projector). Stage-2 (`_s2a.py`, cert machinery byte-verbatim; the S4 readable and full rank-48 fold KL byte-replayed all four sampled cells to the digit) confirms this empirically and rules out a numerical artifact: across R = 20 random orthonormal bases per object, the folded-read reconstruction KL is invariant to **0.0** (fp32), the minimal necessity rank k* is unchanged (shift 0 at all four cells; k* = 16/40/32/24), the certified pass/fail never flips (0/80), and the door/core ablation footprints are invariant to ≤ 3×10^{-5} — while a *matched-random* rank-k subspace fails to reconstruct (KL 0.26–0.38 ≫ floor), so the invariance is object-specific, not vacuous. The one honest qualifier is the **core-field identities**: the 19 fields are a *privileged* orthonormal basis of an intrinsic, rotation-invariant 19-dim subspace — the subspace and its rank are basis-invariant, but a within-span rotation mixes the fields, so the *per-axis* English labels are basis-relative (0/19 of the rotated axes match a frozen field, though the 19 fixed field directions keep their names). We read this as rotation-stable for the k*/necessity/folded-read claims and basis-relative for the field *naming*: the "19 named core fields" are one interpretable labeling of an intrinsic subspace, not a canonical per-axis identity (STAGE2_PROPOSAL item A).

6.2 The grammar (GRAMMAR_TABLE_V1): 36/36 seam-cells linear-certified

At the 19-field core grain, the seam map φ_b → φ_{b+1} was fit with four families (linear ridge, blockwise, MLP, door-mediated) and certified *behaviorally*: inject the predicted next-boundary delta and demand KL ≤ floor. All 36 cells certify LINEAR, all in the TIGHT sub-band (KL_LIN ≤ 0.09355 = floor/2; range 0.0010–0.0785); the linear verdict is regime-stable at 12/12 seams. 34 seams are PROPAGATION (a copy baseline also passes; the linear map refines it) and 2 are REWRITE (embed→L0 in prose and code, where the copy baseline fails and the map earns its keep) — the language has a fixed, simple composition law, with real rewriting confined to its first seam. Per-seam effective rank 4.2–16.9; the loosest corner is the final seam b11→12 (R^2_med 0.645 in code), still TIGHT. The three executable rungs are carried as the repetition-regime seam entries at b5–b7. This is a claim about the model — scoped by the perturbation test below to the propagation seams: the certified fields compose linearly at every depth, with the sole embed→L0 rewrite seam an honest exception.

Perturbation robustness (v1.1; answering review Q5). Because the linear verdict is certified on one *global* 19-field dictionary fit once, we tested (Stage-2, `_s2c.py`, seam-certifier machinery byte-verbatim; the global-core cert byte-replayed all 36 frozen KL_LIN to the digit, max deviation 0.0) whether the nonlinearity could hide in the field *definitions* rather than the seams. Three arms: **(C2a) within-span rotation** of the field basis ($C \to C \cdot R$, $R \in SO(19)$) leaves every cell's KL_LIN exactly unchanged (the write is a projector, basis-identical; max dev 0.0) — the law has no dependence on the choice of orthonormal basis inside the field span. **(C2b) ±ε dictionary jitter** (N=20 at ε=0.05, plus ε=0.02 and 0.10) keeps all 36 cells TIGHT on every one of the 30 draws; the loosest cell (code b11→12) holds at ≈0.078, under the 0.09355 bar. **(C1) independent per-seam re-derivation** — refitting the 19 fields on *each seam's own* boundary-pair residuals (top-19 SVD) — keeps **33/36** cells LINEAR-TIGHT. The three that do not are all the *same* seam, the first (embed→L0), in every regime: prose and code stay LINEAR but lose the TIGHT sub-band (KL 0.108, 0.173), and repetition breaks linearity outright (KL 0.509 > floor). This is exactly the seam singled out above as the language's *only* REWRITE seam. The honest reading: the linear composition law is a robust property of the *model* at all 33 propagation seams — it survives field-basis rotation, ±ε jitter, and fully independent per-seam field re-derivation — while the embed→L0 rewrite seam's linearity is a property of the *certified global fields*: independently-derived first-seam fields expose the detokenization nonlinearity that the global fields linearize. We therefore scope the model-level linearity claim to the propagation seams and label the b0 rewrite seam field-conditioned, as the pre-registered kill branch directed (STAGE2_PROPOSAL item C).

6.3 The inverse (ENCODER_V1): exact, well-posed, and honest about its span

The encoder (gloss → state) is *defined, not trained*: the algebraic right-inverse of each readable channel. Worst relative round-trip residual over all projection channels: 1.81e-4 (bar 1e-3); the door-summary right-inverse lands at 1.6e-6. Encode-then-decode substitution is well-posed at 39/39 cells under the primary meter with zero byte-replay misses (the 3 legacy-only exceptions are exactly the V7 meter-residue cells — an independent cross-check that the encoder reproduces the decoder's per-cell disposition to the digit). Off-span behavior is measured, not assumed: pushing glosses to $|k|=10$ beyond the visited manifold, 5/8 named axes are MANIFOLD-BOUND (indistinguishable from matched noise off-span) and 3/8 show structure (the onset rung R=1.95, the naval core field R=1.58 with directional structure, the clause-boundary corridor axis R=1.52); the DEAF glitch control lands exactly where an inert axis should (R=0.95), validating the instrument.

6.4 The speak test: the language runs both ways

Fig. 5. Three tiers, falsifiers locked before any tier ran:

- **T1 RECONSTRUCT — PASS 39/39.** Read state → English gloss → re-encode → substitute: behavior inside the recalibrated floor at every cell (byte-replaying both the decoder bank and the frozen well-posedness table first; 0 misses). Example, prose b6 at token " km": the decoder reads *mixed-measurement field +3.2σ, clause-final/physical-process +2.5σ*, content image [apiece, increments, km]; re-encoding costs KL 0.146 vs floor 0.187 — the model continues identically. (Stated for the skeptical reader: T1 is close to a corollary of V7's 39/39 plus the exact algebraic inverse; what it adds — and what is tested here — is the *gloss-serialization bottleneck*: the state is rebuilt from the serialized English readout alone, not from cached projections.)
- **T2 TRANSPLANT — PASS, 94.7% vs 18.6%.** Encode context A's readable gloss (content + corridor
- surface term) into context B, keeping B's dark residual: across 16 prose pairs at b6, the transplant closes 94.7% (SE 0.1%) of the B→A next-token behavioral gap; a matched-random readable move of the same norm closes 18.6% (margin 0.761, bar 0.15). Transcript: B = "Ouellette, of H Company, left the perimeter to" predicts [take, join, assist, help, return]; with A's gloss (A = a Battle-of-Yongsan narrative) it moves onto A's distribution [a, the, about, held, being] (s = 0.953). The transferable behavioral meaning lives in the readable subspace; the dark residual does not carry it. *L5-post:* the un-transplanted 5.3% is certified MISSING-MASS — adding the certified door channels to the payload (summarized read_W read or full Q_union) closes exactly 0.0 more, the raw un-summarized door subspace closes only ~27% of the residual, and only the full raw state closes the gap (§6.6). The remainder lives in un-adjudicated dark mass outside the entire certified dictionary; 94.7% is *not* upgradable to "100% of transferable meaning accounted." *L6-post:* that dark carrier is itself now characterized — DIFFUSE (no SVD subspace of the 329-dim dark complement at rank ≤ 256 closes 80% of the transplant gap; the best, k = 256, closes 0.48 while a *random* 256-dim dark slice closes 0.38) and mostly word-less (of its 8 largest carrier directions, 2 cross the naming bar faintly via non-vocabulary channels and 6 are certified no-gloss), §6.7.

Boundary × regime generality (v1.1; answering review Q4). The 94.7% above is one mid-stack boundary (b6), prose, 16 pairs. Stage-2 (`_s2b.py`, T2 machinery byte-verbatim; the b6/prose 16-pair closure byte-replayed to 0.9467, dev 0.0) re-ran the transplant on a 3×3 grid — boundaries {early b2, mid b6, late b10} × regimes {prose, code, repetition}, 32 pairs/cell, matched-random null per pair, 10k-bootstrap CIs. Robust central closure (per-pair *median*, since code is heavy-tailed):

closure (median)	prose	code	repetition
early b2	0.98	0.70	0.98
mid b6	0.95	0.60	0.90
late b10	0.94	0.89	0.82

Two findings. **(i) Not b6-special:** prose closes 0.94–0.98 and repetition 0.82–0.98 at early, mid *and* late boundaries — the "one boundary" caveat is discharged for prose and repetition. **(ii) Regime-sensitive in code at early/mid depth:** the readable-gloss subspace (corridor+fold reader bases, thin in code, §6.1) carries the transferable behavior in code only at late depth (b10 = 0.89, all 32 pairs in [0.86,0.91]); at b2/b6 the code transplant is heavy-tailed — median 0.70/0.60 but a minority of pairs diverge catastrophically, collapsing the *mean* to 0.31/−0.01. All nine cells clear the pre-registered margin band (closure − null > 0.15, bootstrap CI excludes the null), but in the two code early/mid cells that band is driven by an even-more-negative random null, so the closure itself is the governing number. We scope the transplant claim to *boundary-general and prose/repetition- general*, with an explicit code-at-early/mid caveat (STAGE2_PROPOSAL item B). - **T3 HUMAN-EDIT (the crown) — CROWN-STEERABLE, 3/4.** Hand-edit one English field by +3σ, re-encode, substitute, and watch the model's vocabulary: the naval edit raises [amphib, sunk, ashore, reefs, sailed, submarine] (response +2.84 vs matched-random null95 0.53); the clause/delimiter edit raises delimiter/flag/whitespace tokens (+0.54 vs 0.12); the operator/keyword edit raises handle artifacts (+0.58 vs 0.011). The response matrix is diagonally dominant for all three (Fig. 5) — each axis steers *its own* English words, matched-random edits never steer. The executable onset rung does **not** steer through this token-image readout (−0.10, smaller than its spillover) — and is now *certified unusable as a steering lever*: probed in L5 through the channel matched to what it is (repeat-onset probability) plus CH-INT and CH-FIELD, its real-but-weak effect clears no internal readout and does not exceed L5's matched-random edit while a positive-control onset direction does (§6.6); at ±6σ it does not separate from an honest N = 20 null, and at ±3σ it sits within the honest N = 20 floor's own draw-to-draw spread (0.00289 < |A| 0.00387 < 0.00587 across two pre-registered draws, §6.6) while scaling sub-linearly — a gauge, not a lever (§6.6–6.7). One pre-registered auxiliary guard was lost and is logged as a finding: the operator axis, expected to be a manifold-bound negative control, *did* steer on-manifold — manifold-boundedness is an off-span property, not on-manifold inertness. The valid discriminator (matched-random null) held everywhere. *L6-post:* the mechanism of that anomaly is now known — the operator diagonal is read-out ECHO: ~94% of the response is the injected certified-word vector riding additively to the unembedding, and the computed downstream contribution is within its matched null (§6.7). The crown verdict is untouched (its discriminator always held), but "operator steers on-manifold" is a statement about direct read-out, not downstream computation, and no genuine on-manifold control is claimed for that axis.

6.5 The exact translation percentage — three honest numbers

There is no single honest number; hiding any of these would be the fudge the program forbids:

1. **Definitional completion:** 4/4 BABEL cells closed = **100%** of the pre-fixed definition — which counts a certified "this channel carries no word" as a completed translation outcome (it is one: the Rosetta answer "this glyph is decoration").
2. **Behavioral round trip:** reconstruct **100%** (39/39); transplant **94.7%** (the missing 5.3% certified outside the entire certified dictionary, §6.6, and characterized diffuse + mostly word-less, §6.7); human-edit **3/4** named axes (the fourth certified a gauge, not a lever, at both tested doses ±3σ and ±6σ — the full two-null story in §6.6–6.7).
3. **English-vocabulary richness (under the L1 σ-matched-null battery at N = 20): 53.6%** of channels carry an explicit English word at the frozen per-gate rate; **46.4%** are certified word-less *under that battery* (corridor 74.3% / core 100% / folds 48.1%). Under explicit multiplicity control (Appendix B, §6.1) the named fraction is correction-dependent — ≈ 26% at channel-level BH q = 0.05, ≈ 7% per-gate, ≈ 9% strong core no correction removes, ≈ 10% built-in false-named bound — so this number, unlike the two above, is scoped to the battery and is the one most in need of a Stage-2 high-N re-draw to sharpen.

An honest 100%-of-definition, with a named 46.4% certified-word-less remainder and one un-steerable rung. *L5-post:* both of the claim's loose ends are now closed as certified negatives, in the direction the favorite bets did *not* predict — the 94.7% does not rise, and the 3/4 does not become 4/4; each remainder is a certified property rather than an open wound (§6.6). *L6-post:* the two follow-up questions those closures opened (BABEL-OQ-3 and BABEL-OQ-4) are answered as well, again mostly against the favorite bets (§6.7); the three numbers above still do not move. Nothing is hand-waved.

6.6 The two remainders, finished (L5): both certified negatives

§6.4 left two named loose ends. A same-day follow-up stage (L5; pre-registered before any measurement, L4/L1 machinery byte-verbatim, payload-1 required to byte-replay T2's 94.67% to the digit before any verdict — it did) closed both. Neither closed the way the favorite bet predicted.

Arm A — where the un-transplanted 5.3% lives: MISSING-MASS. The T2 transplant was re-run at b6 (same 16 pairs) with nested payloads: readable gloss $\bar{s}$ = 0.9467 (residual KL 0.3004); + certified door-summary read 0.9467 (increment exactly 0.0); + full Q_union 0.9467 (0.0); + raw un-summarized door subspace 0.9612 (~27% of the residual); full raw state 1.0000. The gap-attribution statistic φ = 0.0 → MISSING-MASS. Captured-mass accounting at b6: readable subspace 95.17%, dark complement 4.83%, certified-door increment 0.0% (both certified door forms lie *inside* the readable subspace), raw-door increment 6.37%. Matched-random null per payload 0.186, byte-matching L4. Consequence: the 5.3% of transferable behavioral meaning that the readable gloss does not carry is certified to lie *outside the entire certified dictionary* — named and word-less channels alike — in un-adjudicated dark mass (BABEL-OQ-3: extend the naming battery into that complement — run in L6 and ANSWERED: diffuse, mostly word-less, §6.7).

Arm B — the rung probed through its own channels: CERTIFIED-READ-ONLY. L4's open question was whether the onset rung steers through a readout matched to what it *is* rather than the vocabulary image. L5 edited the rung via ENCODER_V1 (byte-verbatim L4 machinery) and watched the behavioral metric M_onset (mean repeat-token probability; clean 0.9569) plus CH-INT and CH-FIELD. The pipeline was validity-gated first: an independent empirical onset direction moved M_onset above its matched-random null (+0.00329 > null95 0.00235, monotone — a pass, though a marginal one; the whole onset-edit magnitude regime sits near the noise floor). The rung itself: effect real but weak — dose-monotone, sign-stable, $|A| = 0.00388 \geq 2$ SE (0.0012), dose 0.01023 at $\pm 6\sigma$ — yet it does **not** exceed a matched-random edit of the same magnitude (null95 0.00993 at the pre-registered N = 3; re-drawn at N = 20 post-audit, below), clears no internal readout (CH-INT 0.874 ≪ null95 9.849; CH-FIELD structurally degenerate at Dmag 2e-6, logged UNMEASURABLE, not silently passed; CH-WU already diffuse in L4). Verdict: the rung is read-only through *all four* certified channels — a stronger negative than L4's single-readout miss. Stated honestly: it is a weak/diffuse actuator rather than literally inert, and whether a higher-N null at the $\pm 6\sigma$ dose separates its effect from random was left as the named open question (BABEL-OQ-4); L6 then ran exactly that test and answered NO — the rung stays steering-unusable at every tested magnitude (§6.7).

Post-audit null tightening (pre-registered re-draw, 2026-07-06). The $\pm 3\sigma$ verdict null above was N = 3 — disclosed in the pen as high-variance, and flagged by the pre-release audit. A pre-registered N = 20 re-draw of that null under the byte-verbatim construction (harness _l5n20.py; gate: L6's 20-draw null95 of 0.00289 byte-replayed first, dev 0.0000000) returned null95 = 0.00587 — the rung's $\pm 3\sigma$ effect (|A| 0.00387) does **not** exceed it (R = 0.66, the pre-registered BELOW branch; the 65% BEATS bet lost, logged as a finding). Taken together with L6's honest-N re-arm draw (null95 0.00289, which the effect marginally exceeds), the two pre-registered N = 20 draws *straddle* the effect: the $\pm 3\sigma$ onset effect sits within the honest matched-random floor's own draw-to-draw spread — statistically indistinguishable from the floor, in the direction that further weakens any steering reading. This re-draw replaces the N = 3 disclosure; the steering verdict is unchanged in every branch by construction (no internal readout cleared at any dose; sub-linear dose scaling). Texture for future work: at this magnitude regime, null95 at N = 20 is itself draw-dependent by a factor of ~2 — an onset-steering claim here would need N ≫ 20 or a variance-pooled null (Appendix A row A33).

Both favorite bets lost; both losses are logged as findings. The headline numbers of §6.5 are unchanged by construction — L5 sharpened the claim, it did not move it.

6.7 Hunting the residual (L6): the 5.3% is diffuse and mostly word-less, the rung has one

listener, and the last anomaly was echo

L5's two closures opened two follow-ups (BABEL-OQ-3: *what is* the 5.3% dark carrier; BABEL-OQ-4: does the rung's dose-monotone effect beat an honest null at ±6σ), and one L4 anomaly remained unexplained (the operator axis apparently steering on-manifold). A third same-day stage (L6; pre-registered before any measurement, L5/L4/L1/L3 machinery byte-verbatim, all reproduction anchors replayed to the digit, zero instrument discrepancies) closed all three. Of L6's five pre-registered favorite bets, three lost — every loss a certified finding.

Arm A — the 5.3% carrier is DIFFUSE (Fig. 6) with a thin, faint named edge. The transplant-gap residual at prose b6 (payload-1 byte-replayed: $\bar{s}$ = 0.9467; full-state control $\bar{s}$ = 1.0) was SVD-decomposed in the 329-dim orthogonal complement of the readable span (leakage into the span ≤ 0.003). The spectrum is flat (σ_1 718.9 → σ_{256} 119.2 — no cliff), and the pre-registered localization bar (a rank-k subspace closing ≥ 80% of the gap) is never met: closure 0.011 at k = 1, 0.061 at k = 32, 0.333 at k = 128, 0.480 at k = 256; greedy selection of the 8 best directions closes 0.040. The specificity control is the sharp part: a *random* 256-dim slice of the dark space closes 0.376 (draws 0.362–0.386) — the best hand-picked quarter of the dark space beats a random quarter by only ~0.10. Verdict: DIFFUSE — the missing meaning is smeared across the dark complement, not hiding in a low-rank carrier. The naming battery (L1, byte-verbatim, sigma-matched nulls) was then run on the 8 largest dark directions: 2/8 cross the pre-registered NAMED bar (f = 0.25 → NAMED-SOME, against the ALL-DARK favorite), and both cross it *faintly, through non-vocabulary channels* (svd4: prose CH-INT 1.043 > 0.972, code CH-FIELD 0.198 > 0.192, a faint watercourse/terrain flavor; svd7: prose CH-INT 0.943 > 0.921, repetition CH-FIELD 0.388 > 0.347, semantically incoherent across regimes), with no vocabulary gloss (wu_stable false everywhere) and no cross-regime concept; the other 6 are CERTIFIED-NO-GLOSS. They are recorded as two *provisional dark* entries (LEXICON_V4_ADDENDUM.md, evidence hashes per entry) and do **not** enter the certified dictionary. A re-transplant completeness gate passes — carrying the readable gloss plus the rank-256 dark payload lands within the recalibrated floor (residual KL 0.1408 ≤ 0.1871; $\bar{s}$ 0.9723) — with the loud pre-registered caveat that a random 256-dim dark slice *also* clears that floor (0.1718): a 256-dim dark budget suffices; no pinpoint carrier is claimed. Texture: the dark gap deepens with depth (b2 0.016 / b4 0.036 / b6 0.053 / b8 0.054 / b10 0.060). The honest claim sharpens to: *94.7% of transferable meaning lives in the readable/certified subspace; the last 5.3% is diffuse, high-dimensional, mostly word-less dark signal* — not a hidden nameable concept. (A post-BABEL probe, L7, then attacked the same residual with *trained* linear/MLP/bottleneck readable→dark surrogates: all beat their shuffled-target twins, a linear student predicts about half the dark — yet the reconstructed dark closes none of the transplant gap. NOT-COMPRESSIBLE: the predictable half of the dark is not the half that carries the meaning. Next-paper scope; documented in L7_CLOSEOUT.md.)

Arm B1 — the rung has one candidate certified listener: CONCENTRATED. "Read-only" bounds what can be *written into* the rung; L6 asked what *reads out of* it. Injecting the rung's output image at b6 and scanning every certified channel at boundaries b7–b12 with a multiplicity-honest null (max-|z| over the whole channel × boundary grid, N = 12: null95 0.3067), one candidate listener clears — the naval/warship core field at b7 (z −0.3153, a 2.8% margin over the 12-draw multiplicity null; positive control, the naval carry itself, z 3.715). An ablation cross-check at b12 agrees (naval core −0.98, naval door −0.93 lead the table). The rung is not causally inert downstream — one candidate channel consumes its output, on a thin margin; whether that listener is functionally load-bearing is named open texture.

Arm B2 — BABEL-OQ-4 answered NO; the rung and the onset writer are INDEPENDENT. At the ±6σ dose against an honest N = 20 matched-random-edit null: |A_on6| = 0.01023 < null95 0.0126 — the larger dose does not separate from random (the ±3σ re-arm beats its own mag-3 null, −0.00387 vs 0.00289: the onset effect grows *sub-linearly* relative to the random-edit floor, which is why pushing harder buys nothing; the post-audit N = 20 re-draw of the ±3σ null returned 0.00587, putting the effect *inside* the floor's draw-to-draw spread, §6.6). The rung is now certified steering-unusable at every tested magnitude — not separable from the honest floor at ±6σ, inside the floor's own spread at ±3σ, sub-linear in between: a gauge, not a lever. Geometry explains it: the working onset direction v_onset lies 86% inside the certified door space and is near-orthogonal to the rung's rank-1 image (cos 0.179); the rung is not even a clean *reader* of onset (Pearson 0.179 < the pre-registered 0.30 validity bar — the reader-validity failure branch fired as designed, so RUNG-IS-READER was unavailable and no positive was forced; the pre-registration had attached BABEL-OQ-4's "NO" to the RUNG-IS-READER branch, so with FB-D fired the verdict falls to INDEPENDENT and the "NO" rests on the cowriter test not firing). The rung is a faithful readout-scale onset signal on a channel independent of the mechanism that writes onset — a gauge, not a lever.

Arm B3 — the operator anomaly dissolves: LEAKAGE (read-out echo). L4's one unexplained sub-prediction was corr_j17 (the operator axis) steering on-manifold despite an off-span manifold-bound classification. L6 replayed the L4 diagonal to the digit (M_ii +0.5769) and then injected the *same* vector at the final boundary BUS[12] — where nothing downstream can compute on it — measuring the pure read-out echo: M_direct 0.5432, ~94% of the full response; the computed downstream contribution M_comp 0.0336 sits *within* its matched null (null95 0.1646; the echo itself is real, 0.5432 > null95_direct 0.0499). The ±6σ decomposition agrees (1.147 full = 1.086 direct + 0.061 computed). The apparent on-manifold steering was the injected certified-word vector riding additively to the unembedding — no genuine on-manifold control; manifold-bound stands (off-span it is regime-conditional: prose R_k10 0.90 and code 0.86 bound, repetition 1.54 structured extrapolation). The T3 crown is untouched — its matched-random discriminator always held — but the operator row's mechanism is direct read-out, not computation (§6.4).

L6 scoreboard: two favorite bets hit (DIFFUSE, CONCENTRATED), three lost (ALL-DARK, RUNG-IS-READER, MIXED) — all three losses logged loudly as findings. BABEL-OQ-3 and BABEL-OQ-4 are CLOSED; the last L4 anomaly is EXPLAINED; every headline number of §6.5 is unchanged by construction.

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7. Relation to prior and concurrent work

7.0 The claim, and its proof

This paper's claim, in its locked form: **the first complete, certified, bidirectional decode of an entire production language model.** Four properties define it: **(1) whole-model coverage with a priced remainder** — every dimension at every boundary priced against a pre-registered floor, with the untranslated remainder itself measured and certified rather than left dark; **(2) behavioral certification** — every entry a falsifiable mechanical verdict against matched nulls, with "provably word-less" a first-class tested outcome; **(3) a constructive route** — decoder and encoder assembled from the model's own certified channels with an algebraic inverse, so the headline results are claims about the model; **(4) a bidirectional behavioral round trip** — read, re-encode, transplant, and human-edit, scored by what the model does after. Table 1 adjudicates each prior and concurrent line against the four; every row is cited in full in the prose of this section, and partial credit is marked wherever a line provides a piece of an axis.

Table 1 — the prior-art table (— = absent; *partial* = a piece of the axis, stated).

work	(1) whole-model coverage, priced remainder	(2) behavioral certification	(3) constructive route	(4) bidirectional round trip
SAE feature dictionaries + all-units explanation scoring, 2023–26 (Cunningham et al., arXiv:2309.08600; Bricken et al., 2023; Templeton et al., 2024; Bills et al., 2023; Bloom, 2024; Gemma Scope, Lieberum et al., arXiv:2408.05147; Paulo et al., arXiv:2410.13928)	*partial* (all-layer dictionary coverage exists — GPT-2-small residual SAEs (Bloom 2024), Gemma Scope — with per-layer reconstruction/CE pricing, and Bills et al. 2023 score an explanation for every GPT-2 neuron; no per-cell pre-registered pass/fail floors, and the remainder is open "dark matter" (arXiv:2410.14670), not certified or characterized)	— (auto-interp plausibility scoring; cf. arXiv:2602.14111)	— (trained external dictionary)	*partial* (feature-steering demonstrations; no certified English→state→behavior protocol)
LatentQA, 2024 (Pan, Chen & Steinhardt, arXiv:2412.08686)	— (sampled activations)	— (open-ended QA scored on curated evaluations)	— (fine-tuned decoder LLM)	*partial* (gradient-based control through the trained decoder)

work	(1) whole-model coverage, priced remainder	(2) behavioral certification	(3) constructive route	(4) bidirectional round trip
Activation Oracles, Dec 2025 (Karvonen et al., arXiv:2512.15674)	— (sampled activations)	— (oracle task accuracy, not per-channel verdicts)	— (trained generalist verbalizer LLM)	— (read-only)
Predictive Concept Decoders, Dec 2025 (Huang, Choi, Johnson, Schwettmann & Steinhardt, arXiv:2512.15712)	— (sampled activations)	*partial* (trained end-to-end to *predict* behavior; no per-channel certified verdicts, no priced remainder)	— (trained encoder–decoder assistant)	— (read/predict direction only)
Natural Language Autoencoders, May 2026 (Fraser-Taliente et al., Transformer Circuits)	— (sampled activations at middle-to-late layers)	— (learned glosses scored by reconstruction FVE)	— (RL-trained verbalizer/reconstructor pair)	*partial* (round trip scored in activation space (FVE); includes a qualitative behavioral steering demo via reconstructed activations — no matched-null or pre-registered behavioral certification)
Cycle-Consistent Activation Oracles, Mar 2026 (Chalnev, LessWrong)	— (sampled activations, Qwen3-8B)	— (cycle consistency; the author is candid it does not force faithfulness)	— (trained cycle)	*partial* (activation-space cycle)
this work	**39/39 boundary×regime cells at floor; unexplained mass 0.000 nats (primary meter); the 5.3% transplant remainder itself certified diffuse and mostly word-less (§6.6–6.7)**	**351/351 channels adjudicated against sigma-matched nulls; CERTIFIED-NO-GLOSS first-class (§6.1)**	**doors/core/corridor/rungs certified necessary on the model; the encoder is the algebraic inverse (§3.1, §6.3)**	**reconstruct 39/39; transplant 94.7% vs 18.6% null; human-edit 3/4, the fourth certified a gauge, not a lever, at both tested doses (§6.4–6.7)**

Every prior row leaves at least one cell empty; this work fills all four. The adjudications are deliberately generous — where a line supplies part of an axis it is marked as such, and each line is engaged on its own terms below.

Verifiability of the claim itself. The provenance is mechanical, not rhetorical: every band in this paper was locked in an append-only findings pen before the measurement it governs (the pre-registration block behind each number is cited in Appendix A), every verdict-bearing artifact is frozen and sha256-stamped (§10), and every headline number is byte-replayable from those artifacts on one workstation GPU. A reader who doubts any cell of Table 1's last row can re-run that cell: the verdict-bearing harnesses — including _l1.py for the 351/351 adjudication and _l2babel.py for the 36/36 seam law — ship with the release bundle.

Falsifiability of "first." If any prior work is shown to provide all four properties for any model, we will amend this claim; the contact route is the repository issue tracker (github.com/wpferrell/babel-codec-gpt2) or the author's address on the title page. Confidence is stated here as openness, not territory.

The read-direction lineage. Projecting or translating intermediate activations into tokens has a long line: logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2023, arXiv:2303.08112); DecoderLens (Langedijk, Mohebbi, Sarti, Zuidema & Jumelet, Findings of NAACL 2024, arXiv:2310.03686), which lets a decoder cross-attend intermediate encoder layers; Patchscopes (Ghandeharioun et al., ICML 2024, arXiv:2401.06102), which patches representations into inspection prompts; LatentQA (Pan, Chen & Steinhardt, ICLR 2026, arXiv:2412.08686), which fine-tunes a decoder LLM to answer open-ended questions about activations; and ParaScopes (Pochinkov et al., arXiv:2511.00180), which decodes paragraph-scale plans from residual streams with trained residual-stream decoders. The line has since scaled into trained generalist explainers: *Activation Oracles* (Karvonen, Chua, Dumas, Fraser-Taliente, Kantamneni, Minder, Ong, Sen Sharma, Wen, Evans & Marks, arXiv:2512.15674, December 2025) train LLMs to answer arbitrary natural-language questions about another model's activations, and *Predictive Concept Decoders* (Huang, Choi, Johnson, Schwettmann & Steinhardt, arXiv:2512.15712, December 2025) train an encoder–decoder assistant end-to-end to predict model behavior from activations through a sparse concept bottleneck — the nearest prior step toward behavioral grounding, and still a trained external assistant over sampled activations. All of these read; none prices a whole model's state space against a behavioral completeness bar, and none certifies channels as word-less. The SAE feature-dictionary program (Cunningham et al., arXiv:2309.08600; Bricken et al., Transformer Circuits 2023; Templeton et al., Transformer Circuits 2024) is the largest-scale naming effort and Table 1's first row: features are named by auto-interp plausibility and demonstrated by steering, not certified by pre-registered behavioral verdicts, and the dictionary's reconstruction gap is an open "dark matter" problem rather than a priced remainder. The whole-model-coverage precedents in that program are engaged directly: Bills et al. (2023) attach an explanation and a simulation score to *every* neuron of GPT-2 — the field's best-known all-units claim — and EleutherAI scale the recipe to millions of SAE features (Paulo et al., arXiv:2410.13928); Bloom (2024) releases SAEs for every residual-stream layer of GPT-2 small with a substitution CE price, and Gemma Scope (Lieberum et al., arXiv:2408.05147) covers every layer and sublayer of Gemma-2 with per-layer delta-loss pricing. All of these price coverage; none certifies per-channel verdicts against pre-registered floors, and none certifies or characterizes its remainder.

Concurrent work on the translation concept (spring 2026). Anthropic's *Natural Language Autoencoders* (Fraser-Taliente, Kantamneni, Ong, Mossing, et al., Transformer Circuits, May 2026) train a verbalizer/reconstructor pair by RL to explain single residual activations of production Claude models (Opus 4.6, Haiku 4.5/3.5) at middle-to-late layers, scored by fraction of variance explained; they credit the independent *Cycle-Consistent Activation Oracles* (Chalnev, LessWrong, March 2026), which trains a cycle-consistent activation→text→activation loop on Qwen3-8B and is candid that cycle consistency alone does not force faithfulness. These works published the English↔activation translation concept; we do not claim it. The present contribution differs on four axes, stated without chest-thumping: **(a) coverage** — NLA reads sampled activations at middle-to-late layers; this program prices *all* dimensions at *all* 13 boundaries and states the translated fraction cell by cell; **(b) certification vs plausibility** — NLA explanations are learned glosses scored by reconstruction quality; every entry here carries a falsifiable mechanical verdict against sigma-matched nulls, and CERTIFIED-NO-GLOSS is a first-class tested outcome with no analogue in that line; **(c) the constructive route** — their translator is a trained external pair; our decoder/encoder is assembled from the model's own certified necessity channels with an *algebraic* inverse, so the internal-alphabet and seam-linearity results are claims about the model; **(d) the behavioral round trip** — their round trip is *scored* in activation space (FVE), with a qualitative steering demonstration on top (edit the verbalizer's explanation, re-encode, steer, observe the completion change — credited here; it carries no matched-random null and no pre-registered behavioral falsifier); ours is scored in *behavior* throughout — substitution floors, cross-context transplant, and human edits the model obeys in the edited field's own vocabulary, each against matched-random nulls.

The dissociation findings. Our decoupling law (§4.2) has concurrent independent company: *Steerable but Not Decodable* (Nadaf, arXiv:2604.02608, April 2026) reports function vectors that steer where no logit-lens readout decodes — the same steering/decoding split our T16→T18→V2 line measured on certified fields and then quantified as the 1-vs-128 output/object asymmetry; *Sanity Checks for Sparse Autoencoders* (Korznikov et al., arXiv:2602.14111, February 2026) shows trained SAEs barely beating random baselines on interpretability, probing and editing metrics — mainstream doubt about variance-based decomposition to which a necessity-certified, null-gated alternative is one constructive answer; steering-vector non-identifiability (arXiv:2602.06801) argues steering vectors form equivalence classes — our behaviorally certified directions are an identifiability anchor that line lacks. Kin results engaged in the pre-registrations include attention-output SAEs finding induction features (Kissane et al., arXiv:2406.17759), SAE "dark matter" (Engels et al., arXiv:2410.14670), massive activations / attention sinks (multiple lines), dimensional collapse of attention outputs (Wang et al., arXiv:2508.16929), causal dimensionality (Sarkar & Deka, arXiv:2605.08740), certified circuits (Anani et al., ICML 2026, arXiv:2602.22968), the brain-like synergistic core (arXiv:2601.06851), and the subspace-patching illusion (Makelov, Lange & Nanda, ICLR 2024, arXiv:2311.17030), whose methods warning our substitution + matched-null design is built to respect. Induction heads (Olsson et al., 2022) are engaged directly by §5.1's finding that they read, rather than write, the repetition object; superposition (Elhage et al., 2022, arXiv:2209.10652) frames the whole enterprise. The gap each pre-registration claimed was scanned before each run and hardened afterwards (two metadata corrections were found and fixed by the program's own citation-hardening pass; all 2026-dated citations were re-verified against live pages on 2026-07-06).

Circuit-level certification, attribution, and benchmarks (v1.1; added per review). Two of this paper's four axes have direct antecedents outside the read-direction lineage, and we place them honestly against Table 1. *Causal scrubbing* (Chan, Garriga-Alonso, Goldowsky-Dill, Greenblatt, Nanda et al., Alignment Forum / Redwood Research, 2022) shares this program's core commitment — axis (2), behavioral certification rather than interpretive plausibility: it validates a hypothesized circuit by resampling activations under the equivalences the hypothesis licenses and measuring recovered loss. It is the closest prior instrument to our substitution meter, and the substitution + matched-null design here is built to respect the same discipline; it differs on the other three axes — it certifies a *hypothesis about a circuit*, not a whole-model per-channel account, prices no remainder, names no channels, and is read/ablate-only (no bidirectional round trip). *Direct Logit Attribution* and its efficient variants — DLA from the mathematical framework for transformer circuits (Elhage et al., 2021), attribution patching (Syed, Rager & Conmy, arXiv:2310.10348, 2023) and automated circuit discovery (Conmy et al., NeurIPS 2023, arXiv:2304.14997) — decompose a logit or a metric into per-component contributions using the model's own components (partial axis (3)), but are read-only linearizations without pre-registered per-channel null-gated verdicts, a priced remainder, or an encode direction (axes (1), (2), (4) absent). *Mechanistic-interpretability benchmarks* — compiled-transformer ground truth (Tracr, Lindner et al., NeurIPS 2023, arXiv:2301.05062), concept-disentanglement evaluation (RAVEL, Huang et al., 2024), and the emerging standardized suites — supply external yardsticks this work does not use: our "ground truth" is instead the model's own behavioral substitution floor, which is why the account is whole-model and self-anchored rather than task-scored. None of these lines provides the whole-model coverage + priced remainder (axis 1) or the bidirectional behavioral round trip (axis 4); each is engaged here for the axis it does share, and the standing commitment of §7.0 to amend "first" applies to them equally.

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8. Limitations

1. **One model, one size.** Everything here is GPT-2 124M. The one cross-model probe in the record points *against* free transfer: the certified late-tail residue has zero direction overlap with Pythia-70m/160m (Jaccard 0.014, p = 1.0) — that residue is GPT-2-specific implementation. What transfers is the *method*; whether the laws transfer is H-GPT2-LAW-VS-RESIDUE (§9).
2. **Boundary grain.** The account prices residual states at layer boundaries. Within-block mechanisms enter only as attribution evidence (§5.1); "complete" does not mean circuit-complete.
3. **Regime scope.** Three regimes. They were chosen adversarially (repetition is where the model is most fragile and was hardest to close), but the account is regime-indexed by construction — the program's own width law (door width 128→446 across distributions at fixed N) says new distributions get their own bill.
4. **The certified-no-gloss fraction is battery-scoped (v1.1).** Every "word-less" claim in this paper means *no gloss under the CH-WU / CH-INT / CH-FIELD battery* — a channel's own unembedding-vocabulary image, its logit-lens contrasts at downstream boundaries, and the antisymmetry of its downstream field response, each against

σ-matched nulls (B_NULL = 12–20), requiring ≥ 2-of-3-regime stability. "Certified word-less" is therefore "un-nameable *under this battery at this budget*," not "provably carries no representational content." A failure to name may reflect the battery's sensitivity/scope rather than semantic absence; plausible battery *extensions* that could name part of the remainder — and are named here as future work (§9, BABEL-OQ-2) — include syntax-tree / dependency probes, morphological classifiers, non-vocabulary (feature-space) linear probes, cross-layer causal-scrubbing readouts, and sparse-autoencoder decoders. Two facts bound the concern: (i) the multiplicity re-analysis (Appendix B) shows the battery is, if anything, *generous* — many marginal gate-clears do not survive an explicit FDR — so the no-gloss fraction is more likely an *under*- than an *over*-estimate of genuine word-less channels; (ii) the L6 dark-complement adjudication (§6.7) shows the battery's honest edge on the hardest mass — 2/8 top dark carriers clear the bar, faintly, without a stable gloss. Whether a richer battery or larger N names some of the remainder is exactly BABEL-OQ-2 (Stage 2). Most of the current no-gloss mass sits in thin code-regime folds.

5. **Meter conventions.** The primary floors are norm-relative by a ratified, sanity-gated convention; the stricter legacy meter is reported permanently alongside (36/39, all three residual gaps priced geometry). The ≤64 unnamed-dims allowance and the prose floor convention (0.1871 at every boundary) are program definitions, written before the data and never moved. *Sensitivity (v1.1, Appendix C):* the 36/39 legacy closure is floor-construction-robust; the 39/39 is reached only at the full norm-relative recalibration (β = 1) and reopens to 36–37/39 under any partial norm-scaling or a 10 % floor tightening — so the norm-relative meter is load-bearing for exactly the last three cells, which are labelled meter corrections, not model discoveries.
6. **Trained rungs.** The three executable rungs are trained modules; the claim is behavioral substitutability at the model's own floor on held-out content under a memorization falsifier — not source-code identity with the circuit. The b5 margin is thin-but-stable (median 7.6% under floor across 80 fresh blocks; worst batch 1.05×) and is labeled so wherever it appears.
7. **The rung is certified unusable as a steering lever at every tested magnitude** (§6.6–6.7): the one crown family that resists human editing. The rung-native probe (BABEL-OQ-1) was run in L5 and answered NEGATIVE across the matched behavioral channel, CH-INT, and CH-FIELD; the ±6σ dose question (BABEL-OQ-4) was run in L6 at an honest N = 20 and answered NO (|A_on6| 0.01023 < null95 0.0126), while at ±3σ the effect sits within the honest N = 20 floor's own draw-to-draw spread (L6 re-arm 0.00289 < |A| 0.00387 < post-audit re-draw 0.00587) with sub-linear scaling — a gauge, not a lever, in both regimes. Three honest notes remain: the L5 positive-control gate passed marginally (0.00329 vs null95 0.00235 — the entire onset-edit magnitude regime sits near the noise floor); the L5 ±3σ verdict null was N = 3 (pen-disclosed as high-variance; retired by the pre-registered N = 20 re-draw, §6.6); and "steering-unusable" bounds what can be written *into* the rung, not what reads out of it —

one candidate certified channel (the naval/warship core at b7, 2.8% over its 12-draw multiplicity null) consumes its output (§6.7).

8. **Single-GPU replication — a strength stated plainly.** Every number in this paper was produced on one 20 GB workstation GPU in runs of minutes to ~1.4 hours, from deterministic streams, with every artifact hashed. There is no scale barrier between a skeptical reader and any row of the account.
9. **Stage-2 robustness scope (v1.1).** Three reviewer-raised robustness questions were run under Stage-2 authorization, each byte-replaying the frozen number to the digit first. (i) *Rotation/basis (Q3, §6.1):* the necessity certificates — folded-read k*, pass/fail, reconstruction — are basis-invariant (k* shift 0, 0/80 pass-flips, a matched-random reconstruction discriminator separating), but the 19 core-field *per-axis* labels are basis-relative — a privileged labeling of an intrinsic, rotation-invariant 19-dim subspace, now stated as such. (ii) *Transplant generality (Q4, §6.4):* boundary- and prose/repetition-general (median closure 0.82–0.98 across early/mid/late depth), but heavy-tailed in code at early/mid depth (median 0.60–0.70, mean collapsed by outlier pairs), clean in code only at late depth. (iii) *Seam linearity (Q5, §6.2):* robust to field-basis rotation, ±ε dictionary jitter, and independent per-seam field re-derivation at all 33 propagation seams; the single embed→L0 *rewrite* seam is field-conditioned. Each scoping is stated in the body; none overturns a frozen headline.

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9. Future work

- **The transfer question (H-GPT2-LAW-VS-RESIDUE).** Re-run the account on GPT-2 medium: which parts are laws (decoupling, surface-coded dark mass, seam linearity, linear-rung learnability) and which are 124M residue? The width-law arithmetic prices the re-certification.
- **The dark remainder, post-L6.** BABEL-OQ-3 (name the 5.3% carrier) and BABEL-OQ-4 (the rung at ±6σ) were both run in L6 and ANSWERED (§6.7) — the carrier is diffuse and mostly word-less; the rung is steering-unusable at every tested magnitude — so neither is future work anymore (nor is the rung-native speak test, BABEL-OQ-1, answered NEGATIVE in L5). What remains is texture: do the two faint provisional dark signatures (LEXICON_V4_ADDENDUM.md) resolve under a richer readout, or are they borderline battery positives at f = 0.25? Is the rung's single certified listener (the naval/warship core at b7) functionally load-bearing? (A post-BABEL probe, L7, has additionally certified the diffuse residual NOT-COMPRESSIBLE from the readable subspace by trained linear/MLP/bottleneck surrogates, located the onset writer as broadly distributed (top-5 components carry 65%), and found NO certifiable onset lever at any dose — next-paper scope, documented in L7_CLOSEOUT.md.)

- **Naming the remainder (BABEL-OQ-2).** Re-run L1 at larger N / richer readouts on the 154 certified-no-gloss folds; how much of the 46.4% is genuinely word-less vs budget-bounded?
- **Multiplicity re-draw and floor CIs (v1.1; the review's statistical asks, Stage 2).** A high-N (≥ 1000) σ-matched-null re-draw of the whole L1 battery would replace the tail-model extrapolation of Appendix B with assumption-free per-gate p-values and a definitive FDR; a GPU re-run emitting per-token substitution KL would give the bootstrap floor CIs of Appendix C. Both are specified with cost, bands, and kill conditions in STAGE2_PROPOSAL.md — Will greenlights per item.
- **Battery extensions for the no-gloss remainder (§8.4).** Syntax-tree / dependency probes, morphological classifiers, non-vocabulary feature-space probes, and SAE decoders could name part of the certified-word-less set; each is a candidate battery channel to add before re-pricing the no-gloss fraction.
- **Wider margins where the paper wants them.** KL-native rung training at b5; drop-s ablations at b6/b7 to finish the input-contract story.
- **Program linkage.** The dead-capacity/suppressor-surgery line (CCT) and the frontier queue (FRONTIER_QUEUE) supply the next pre-registered targets; the discipline of §2 carries unchanged.

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10. Artifact and reproducibility statement

All verdict-bearing artifacts are frozen, hash-stamped (sha256[:16]), and **publicly released**: the complete frozen bundle is deposited at Zenodo, DOI [10.5281/zenodo.21230108](https://doi.org/10.5281/zenodo.21230108) (versioned; concept DOI 10.5281/zenodo.21230107 always resolves to the latest), with the verdict-bearing harnesses and a mirror of the bundle at github.com/wpferrell/babel-codec-gpt2 and huggingface.co/wpferrell/babel-codec-gpt2. This release answers review Q8 in full: code, exact deterministic streams, and every frozen artifact (not only hashes) are available for independent byte-replay. The full list with the number-to-source trace is Appendix A. Headline artifacts: the standing decoder decoder_v7_tensors.pt `b1d2f464c00c3ef6` / decoder_v7.json `b55e7df9b9e0d9f7`; the primary meter _v5_floors_recal.json `71549ae3afcc8d07`; LEXICON_V3.md `71a51619a9bb25c3`; GRAMMAR_TABLE_V1.json `da6f8a63a061782b` with seam operators _l2babel_maps.pt `b43f877af68728df`; ENCODER_V1 (_l3_encoder.pt `6be189567c41e91d`, ENCODER_V1.json `365dc3ff592fc6bd`) with WELLPOSEDNESS_TABLE_V1.json `ea5236cbd608a385` and OFFSPAN_TABLE_V1.json `77dd0948a63bb24f`; the speak-test record _l4_result.json `d777297d1fae0e92`; the L5 remainder-closure record _l5_result.json `14c3e0db902824dd` (harness _l5.py `982cf19ab4647e81`; prose records L5_CLOSEOUT.md, BABEL_CLOSEOUT_ADDENDUM.md); the L6 dark-mass/rung/anomaly closure record _l6_result.json `fd71cd4f3bbfa3e3` (harness _l6.py `d7ab446ba5aacaa5`, bases _l6_bases.pt `a60a0ab67b85c410`; prose records L6_CLOSEOUT.md, BABEL_CLOSEOUT_ADDENDUM2.md, LEXICON_V4_ADDENDUM.md). Pre-registrations and results live in an append-only pen (FINDINGS_PEN_CONSTRUCTIVE_2026-06-28.md) whose block headers are cited throughout Appendix A. Figures 1–6 regenerate from the frozen JSONs alone via paper_figs/make_paper_figs.py (CPU, matplotlib). Compute: one NVIDIA RTX A4500 (20 GB), fp32, eager attention, TF32 off.

Appendix A — Number provenance (every headline number → its frozen source)

Provenance convention: the pen is append-only; for the L2–L5 stages, whose results headers carry date-only stamps, append order is the pen's clock.

#	number(s)	source (artifact :: field, or pen block)
A1	39/39 recal, 36/39 legacy, 0.000/0.283 nats, N_open 6/39	_v7_result.json :: verdict.tables; pen "V7 — RESULTS (2026-07-05 ~23:10)"
A2	11.18 nats/19 cells (campaign verdict)	_open6_result.json :: gap_table (sum); OPEN_CAMPAIGN_CLOSEOUT.md §5
A3	9.83 (V2 S5), 7.549/17 (V3, V4), 6.819/14 & 7.172/17 (V5), 3.125/11 & 3.402/14 (V6)	_v2/_v3 result gap_table sums; _v4_result :: verdict; _v5/_v6_result :: verdict.tables
A4	birth cells: rep b1 3.448→0.061 vs floor 0.135	_open6_result.json :: cells; pen OPEN-6 RESULTS
A5	rooms: BUS[2] 4.84 dark nats; g_room 0.8614 (0.722–0.966)	pen OPEN-1 RESULTS (H-OPEN1-c); pen OPEN-6 RESULTS (H-OPEN6-b)
A6	rooms dark to doors 0.231–0.410 vs bar 0.50; EMB-CUR 0.416–0.761; MLP share 0.833–0.877	pen OPEN-3 RESULTS (H-OPEN3-a/-c, FINDING 1)
A7	wte scope null: rep b12 field-R^2 0.563→0.513	pen OPEN-6 RESULTS (ladder texture)
A8	outlier oracle: −0.13 median; rep_b11 +0.24 anomaly; poisoned dim 1; rep_b12 field-R^2 0.59→0.84 for −0.24 nats	V2_CLOSEOUT §1; V3_CLOSEOUT §1 (armB0); _v2/_v3_result.json
A9	rep-b5 character: 5.6× rep-specific, 159-dim, L2-read R^2 0.55 (banked 0.5525), token −0.22, ablation 2× exit / 15× null	V2_CLOSEOUT §1; _v2_result.json (armB/armC)
A10	circuit: MLP0 77.6% energy, 5.3× specific, exit 0.045→~1.0; heads a0.h1/a0.h5, a1.h7/a1.h4/a2.h10; induction heads downstream	V3_CLOSEOUT §1; _v3_result.json :: armA
A11	S8m null: 4.256 vs wall 3.460, 15/20 nulls; v1r byte-identical 4.25577; v3 co-add 3.41905 (14/20); full width 2.579/2.333	V4_CLOSEOUT §1; V5_CLOSEOUT §1; _v4/_v5_result.json
A12	k*(m0 rep) = 1: V4 margin 1.7% → V5 63.5% (dev 0.18404 ≤ 0.50466); k_obj* ≈ 128	V4_CLOSEOUT §1; V5_CLOSEOUT §1 (OQ-V4-1)
A13	recal floors: wall bars moved <9% (0.1223→0.1279 / 0.0777→0.0806 / 0.0997→0.1072); 26 legacy floors replicated; 3 cells re-priced	V5_CLOSEOUT §1; _v5_floors_recal.json

#	number(s)	source (artifact :: field, or pen block)
A14	V6 rung: SACRED 0.11172 ≤ 0.1279; R^2 0.77; twin 1.53558; within 0.013; MLP/ATTN 0.354/0.345 (train-R^2 0.978/0.967); HOLD2 0.1317; copy fidelity 0.9636; loss curves	_v6_result.json :: ladder, cert.rungs; V6_CLOSEOUT §3
A15	C1 discharge: median 0.11814, 4/5 under floor, worst 1.05×; b6 0.02644 ≤ 0.08057 (0.9995); b7 0.06665 ≤ 0.10724 (0.9711); drop-s 0.11105	_v7_result.json :: c1, armB, c2; V7_CLOSEOUT §1/§3
A16	code column folds: all 9 close both meters; thinnest code_b5 4.7%; allowance 61–62 ≤ 64	_v7_result.json :: armA + verdict tables; V7_CLOSEOUT §1
A17	corridor naming arc: W_U 2/35; field 33/35 → sigma-honest 19/35, BROAD-SURVIVES 21/35; r_var 0.11, r_causal 0.19	V3_CLOSEOUT §1 (armC); V4_CLOSEOUT §1 (V4C)
A18	LEXICON_V3 counts: 35/19/297; 188/163 = 53.6%/46.4%; corridor 26/35; folds 143/297 (15 regime-specific); per-cell NAMED/DONE splits; glitch DEAF (field 0.87 vs null95_D 118)	LEXICON_V3.md `71a51619a9bb25c3`; L1_CLOSEOUT §1/§4 (the glitch-field 0.87 vs null95 ≈ 118 and corridor 26/35 numbers are prose-sourced there — no JSON field); pen L1 RESULTS (2026-07-06)
A19	grammar: 36/36 LINEAR TIGHT (≤0.09355); KL_LIN 0.0010–0.0785; 34 PROP/2 REWRITE (b0 prose 0.2002 / code 0.2772); R^2_med 0.645–0.949; rank_eff 4.2–16.9; 12/12 regime-stable	GRAMMAR_TABLE_V1.json `da6f8a63a061782b`; L2_CLOSEOUT §1/§4
A20	encoder: M1 worst 1.81e-4 (channels 3.0e-7…1.1e-4; read_W 1.6e-6); M2 39/39 recal (36/39 legacy, same 3 cells), KL 0.019–0.366; M3 5/8 MANIFOLD-BOUND, rung R=1.95 / naval R=1.58 / clause R=1.52, DEAF control R=0.95	_l3_result.json; WELLPOSEDNESS/OFFSPAN tables; L3_CLOSEOUT §1
A21	speak test: T1 39/39 (0 misses; KL 0.019–0.3665; legacy 36/39); T2 $\bar{s}$ 0.9467 ± 0.001 vs null 0.186 (margin 0.761, 16 pairs); T3 matrix rows +2.838/+0.542/+0.577/−0.104, null95 0.532/0.116/0.011/0.088; demo transcripts verbatim	_l4_result.json `d777297d1fae0e92`; pen L4 RESULTS; BABEL_CLOSEOUT §3–4
A22	in-flight demo: P1 3/3; P2 R^2 0.5016; P3 +0.2168, SE 0.0645 (3.4 SE), anchor 0.22181	_open6_result.json :: a3, a3_verdict; NARRATED_GENERATION_V1.md
A23	instrument genealogy: T6 floor 0.1871 (EPS_MSE 0.080085); T7 k=*13 front door, interior k*=1; T9 attn 128/MLPI 240/union 385; T11 widths 128/163/279/446; T14b 19-dim core; T15 19/19-17/19-0/19; T16 OPAQUE 1/19	pen T6/T7/T9/T11/T14/T15/T16 blocks; PLAYBOOK ch.12 (conventions verified on disk)
A24	residue transport zero: Jaccard 0.014, p = 1.0; top-5 dirs 38–64% late-tail variance, MP-excess 38–107×	OPEN_CAMPAIGN_CLOSEOUT §1 (OPEN-5 J5)
A25	rep floor split: early 6.4× survives norm-match (2.16 nats); late ≈ 0.194 ~ prose (geometry)	V3_CLOSEOUT §1 (armD)

#	number(s)	source (artifact :: field, or pen block)
A26	single-GPU: RTX A4500; round GPU times 183.6 s (OPEN-6), 373.8 (V2), 947.9 (V3), 4891.6+712.1 (V4+V4C), 4262.8 (V5), 1821.0 (V6), 1553.7 (V7); BABEL codec 4647.3 (L1 leg 3), 3172.5 (L2), 129.6 (L3), 74.9 (L4)	result JSONs :: elapsed_s; closeout OPS lines; pen OPEN-5 RESULTS (GPU model)
A27	citations §7	live-page fetches 2026-07-06 (NLA, CCAO, LatentQA, ParaScopes, DecoderLens, 2604.02608, 2602.14111) + pen CITHARD block (2026-07-03)
A28	L5 Arm A: ladder 0.9467 / +cert-door 0.9467 (+0.0) / +Q_union 0.9467 / +raw-door 0.9612 (~27% of residual) / full 1.0000; φ = 0.0 (MISSING-MASS); mass 95.17/4.83%, cert-door inc 0.0%, raw-door inc 6.37%; null 0.186	_l5_result.json :: armA `14c3e0db902824dd`; pen "L5 — RESULTS (2026-07-06)"; L5_CLOSEOUT §3
A29	L5 Arm B: clean M_onset 0.9569; pos-ctrl +0.00329 > null95 0.00235; rung A_on −0.00388 (2 SE 0.0012; sign-repro; A_on6 −0.01023) vs matched-random null95 0.00993 → NO-STEER; CH-INT 0.874 ≪ 9.849; CH-FIELD Dmag 2e-6 UNMEASURABLE → CERTIFIED-READ-ONLY	_l5_result.json :: armB `14c3e0db902824dd`; pen "L5 — RESULTS (2026-07-06)"; L5_CLOSEOUT §3
A30	L6 Arm A1/A3: ladder closure k1 0.0111 / k32 0.0606 / k64 0.1895 / k128 0.3331 / k256 0.4795 (no k ≤ 256 ≥ 0.80 → DIFFUSE); greedy-8 0.0399; random-256 closure 0.3763 (draws 0.3617/0.3857/0.3814); svals 718.86 → 119.2; dark→span leak ≤ 0.00299; re-transplant s̄_sel 0.9723, residKL_sel 0.14078 ≤ 0.1871 PASS (caveat: random-256 residKL 0.17184 also clears); per-boundary dark gap b2 0.0161 / b4 0.0360 / b6 0.0533 / b8 0.0544 / b10 0.0600	_l6_result.json :: armA `fd71cd4f3bbfa3e3`; pen "L6 — RESULTS (2026-07-06)" (pen 9473–9613); L6_CLOSEOUT §3
A31	L6 Arm A2: 8/8 dark dirs adjudicated, 2 NAMED (f = 0.25, NAMED-SOME), both non-WU: svd4 prose CH-INT 1.0426 > 0.972 + code CH-FIELD 0.1984 > 0.1921 (watercourse/terrain flavor); svd7 prose CH-INT 0.9425 > 0.9206 + rep CH-FIELD 0.3881 > 0.3466 (incoherent); 6 CERTIFIED-NO-GLOSS; provisional entry hashes 05bca41921237c1c / 6e787a28cc4492d9	_l6_result.json :: a2_words/a2_verdict; LEXICON_V4_ADDENDUM.md (the two entry hashes are per-entry canonical-record hashes living there, recomputable via json.dumps(record, sort_keys=True, separators=(',',':')) — not string fields of _l6_result.json)
A32	L6 Arm B: B1 one listener — naval/warship core @ b7 z −0.3153 vs multiplicity-honest null95_max 0.3067 (N = 12; pos-ctrl naval carry 3.715; ablation b12 −0.9768/−0.9329) → CONCENTRATED; B2 cos(v_onset, rung_img) 0.1786, v_onset span5 frac 0.8612, reader r 0.1789 < 0.30 (FB-D fired), A_on6 −0.01023 < null95_20_mag6 0.0126 (±3σ re-arm −0.00387 > 0.00289) → INDEPENDENT, BABEL-OQ-4 NO; B3 M_ii 0.5769 = direct 0.5432 + comp 0.0336	_l6_result.json :: armB `fd71cd4f3bbfa3e3`; pen "L6 — RESULTS (2026-07-06)"; L6_CLOSEOUT §3

#	number(s)	source (artifact :: field, or pen block)
	(null95_comp 0.1646, null95_direct 0.0499; ±6σ 1.1467/1.086/0.0607) → LEAKAGE; off-span R_k10 prose 0.9024 / code 0.8585 / rep 1.539	
A33	L5N20 post-audit null re-draw (±3σ, N = 20): GB-1 replay A_on −0.00387 / mag3 7.8544 / M_clean 0.95686; M3 null-machinery replay null95 0.00289 (dev 0.0000000 vs L6's banked null95_20_mag3); fresh draw (seed 20260745) null95_20 0.00587 (draws 0.00020–0.00610, mean 0.00238); R = 0.00387/0.00587 = 0.6593 → BELOW (the 65% BEATS bet lost, logged); the two N = 20 draws straddle the effect (0.00289 < 0.00387 < 0.00587)	_l5n20_result.json `f937ba7caf0f5cc6` (harness _l5n20.py `524cb4004abebe44`); pen "L5N20 — GAP + PRE-REGISTRATION (2026-07-06 ~19:03)" + "L5N20 — RESULTS (2026-07-06 ~19:11)"

Frozen-artifact hash table (verdict-bearing set): decoder_v7 `b1d2f464c00c3ef6`/`b55e7df9b9e0d9f7`; _v7_result `e293979a6be37109`; _v7_bases `3165bc033ffaec6d`; _v6_result `dcb767be66434df3`; decoder_v6 `a2d384d29c27fb91`/`92d57bd895b4fe9f`; _v6_bases `49eda0c1a24bbfce`; _v5_result `67d37d8ec1b1abb3`; _v5_floors_recal `71549ae3afcc8d07`; _v5_bases `04c401d24ab2cd9d`; _v4_result `cf5e67d2057c2ad0`; _v4c_result `e5362ff0aa55ca72`; _v4_bases `5fe42657e04b2117`; decoder_v3 `296dcacca0102cec`/`ac38852bfdab64d7`; decoder_v2 `f380199539175b32`/`d36a32a9eaaf206f`; decoder_v1 `0f47849983d95f80`/`868f15c41bf593d5`; LEXICON_V3 `71a51619a9bb25c3`; _l1_result `52550bc94466a53e`; _l1_bases `8b2d9f72e811af92`; GRAMMAR_TABLE_V1 `da6f8a63a061782b`; _l2babel_maps `b43f877af68728df`; _l2babel_result `b61a144400bab79c`; _l3_encoder `6be189567c41e91d`; ENCODER_V1.json `365dc3ff592fc6bd`; WELLPOSEDNESS_TABLE_V1 `ea5236cbd608a385`; OFFSPAN_TABLE_V1 `77dd0948a63bb24f`; _l3_result `6098d8459eddb360`; _l4_result `d777297d1fae0e92`; _l4_bases `6b557828e949e60d`; _l5_result `14c3e0db902824dd`; _l6_result `fd71cd4f3bbfa3e3`; _l6_bases `a60a0ab67b85c410`; figure inputs not already listed: _open6_result `f4ee0bafd167c377`, _v2_result `aebce3b001cfc304`, _v3_result `b57c38a643b5fdcc`; the post-audit null re-draw _l5n20_result `f937ba7caf0f5cc6`; harnesses _v6.py `cd81a336b4c86a3d`, _v7.py `3e76e505bebb8348`, _l3.py `19455ebfc702b8da`, _l4.py `36d0fbbc8a2f0e3f`, _l5.py `982cf19ab4647e81`, _l6.py `d7ab446ba5aacaa5`, _l5n20.py `524cb4004abebe44`.

Appendix B — Multiplicity control on the naming battery (v1.1; review Q2)

Pre-registered before computing (pen block "REVISION-FDR — PRE-REGISTRATION (2026-07-08)"); harnesses `_rev_fdr.py`, `_rev_fdr_channel.py`; frozen input `_l1_result.json` `52550bc94466a53e`. Downgrade/annotate-only — the frozen L1 verdicts remain the record.

Setup. Each naming gate `(channel, test ∈ {CH-WU, CH-INT, CH-FIELD}, regime ∈ {prose, code, repetition})` stores an observed statistic and `null95` = the 95th percentile of a 20-draw σ-matched null; the raw draws are not frozen. A channel is NAMED when a gate clears (`obs > null95`, plus a ±1σ/±2σ dose-consistency filter) in ≥ 2 of 3 regimes. Family = all 2592 gates on the 312 channels with frozen per-gate stats (297 folds + 15 corridor directions; frozen NAMED = 149/312 = 47.8 %). **Validation gate (passed):** re-deriving NAMED from the raw statistics with the frozen `obs > null95` rule reproduces the frozen verdicts channel-for-channel — 0 mismatches / 312 — so the correction below swaps only the gate for BH-significance.

Continuous p-value. Only one null quantile is frozen, so a continuous p needs a tail model. We anchor a half-normal(μ = 0) null to that quantile so `obs = null95 ⇔ p = 0.05`: `p = 2·(1 − Φ(Φ⁻¹(0.975)·obs/null95))` (primary), with a normal(μ = 0) sensitivity `p = 1 − Φ(Φ⁻¹(0.95)·obs/null95)`. NAMED is re-derived keeping every frozen dose/sign/SE filter and replacing `obs > null95` with "gate survives BH."

Results.

control (312 channels; frozen NAMED 149 = 47.8 %)	q = 0.01	q = 0.05	q = 0.10
per-gate BH, half-normal (m = 2592; **strict, pre-registered**)	6 (1.9 %)	**21 (6.7 %)**	51 (16.3 %)
per-gate BH, normal (sensitivity)	1 (0.3 %)	9 (2.9 %)	21 (6.7 %)
channel-level BH, conjunction-aware (m = 312; **post-hoc**)	28 (9.0 %)	**80 (25.6 %)**	111 (35.6 %)
Holm–Bonferroni FWER, per-gate (α = 0.05)	—	1 (0.3 %)	—

Global-null bound (assumption-free): under H0 the ≥ 2-of-3-regime rule at its design gate α_0 = 0.05 has an expected false-NAMED count of 15.5/312 → false discovery proportion **≈ 10 %** vs the 149 observed names (the dose filter only lowers it). Of the 149 frozen names, 126 have channel p ≤ 0.05, 70 ≤ 0.01, 28 ≤ 10^{-3}, 16 ≤ 10^{-4}.

Reading. Per-gate BH is a *lower bound*: it treats a channel's nine gates as nine hypotheses and discards the ≥ 2-of-3-regime conjunction the rule requires. Channel-level BH credits that conjunction and is the natural unit for "how many channels are named." Most frozen names rest on individually-marginal gate clears made credible by cross-regime replication, so ≈ half survive channel-level BH at q = 0.05 and a ≈ 9 % core survives at q = 0.01. **Binding limit:** the N = 20 null floors the assumption-free empirical p at ≈ 1/21 = 0.048, so continuous p below ≈ 0.05 are tail-model extrapolations; a definitive FDR requires a Stage-2 high-N (≥ 1000) σ-null re-draw (STAGE2_PROPOSAL.md). **Scope:** the 19 core fields (all named, adjudicated in T15) and ≈ 20 further corridor directions (V3/V4C) lie outside this file, so the corrected overall 351-fraction is ≈ 23 % (out-of-file names uncredited) to ≈ 34 % (credited) at channel-level BH q = 0.05, versus the frozen 53.6 %.

Appendix C — Floor-construction sensitivity for the 39/39 closure (v1.1; review Q1)

Harness `_rev_floor.py`; frozen inputs `_v7_result.json` (verdict.tables) + `_v5_floors_recal.json` (legacy + recal floors, per-cell RMS norms). Per-cell mean substitution KL only — per-token KL is not frozen, so a bootstrap CI over holdout tokens is deferred to Stage 2.

Recalibrated floor construction: `floor_recal(b,reg) = mean_4dirs KL(m_rel · noise)`, `m_rel = m*(b)·rms(b,reg)/rms(b,prose)`. The legacy floor is prose-anchored (0.1871) with per-cell measured code/rep floors.

Frozen closure: legacy 36/39, recal (primary) 39/39.

Fractional norm-scaling sweep `floor_β = floor_legacy·(floor_recal/floor_legacy)^β` (log-interpolation through the two measured endpoints; β = 0 legacy, β = 1 the adopted recal):

β	0	0.25	0.5	0.75	1.0	1.25	1.5
closure /39	36	36	36	37	**39**	39	39

The 39/39 is reached **only at the full norm-scaling β = 1**; the last three cells reopen under any partial scaling. Linear interpolation legacy↔recal agrees (36 up to λ = 0.5, 37 at 0.75, 39 at 1.0).

Robustness to a uniform bar shift (multiplier c on the recal floor): 39/39 for c ≥ 1.0; 34/39 at c = 0.9; 26/39 at c = 0.8 — the closure tolerates loosening, is sensitive to ~10 % tightening on ~5 cells. **Margin bands (recal):** 14 cells KL/floor ≤ 0.5, 12 in 0.5–0.8, 13 in 0.8–1.0, 0 fail (legacy: 12 / 13 / 11 / 3-fail).

The three legacy-only gap cells (close only under recal): code_b12 KL 0.2701 vs legacy 0.1398 (×1.93 fail) → recal 0.3589 (×0.75); rep_b12 0.1815 vs 0.0706 (×2.57) → 0.1935 (×0.94); rep_b11 0.1316 vs 0.0894 (×1.47) → 0.1377 (×0.96). Each floor ratio (2.57 / 2.74 / 1.54) tracks the squared regime/prose RMS ratio ρ^2 (2.48 / 2.40 / 1.40) — the documented late-layer norm-geometry effect (§5.1), a genuinely meter-dependent closure labelled as such, not new model behavior. **Honest headline: 36/39 floor-construction-robust; 39/39 under the pre-registered norm-relative meter at full norm-scaling.**

Appendix D — Student (executable-rung) training and validation (v1.1; review Q6)

From `_v6_result.json` (cd81a336b4c86a3d), `_v7_result.json`, harnesses `_v6.py`/`_v7.py`. Every number verified against the frozen JSON.

Task / inputs. Predict the BUS[5] repetition object (span5-complement) from readable inputs only, feature dim 1537 = `x2` (768-dim layer-2 state, centered) ⊕ `ecur` (768-dim current-token WTE embedding) ⊕ `s` (1 scalar, m0's k* = 1 coefficient). Substitute at BUS[5]; meter on never-seen repeat periods.

Data / splits (period-64 repeated segments). 96 training periods (distinct 512-token blocks, seeds 7000–7095) → 30,720 rep-era training tokens; scaler fit from train positions [64, 384) only. Held out: `SACRED` 16 periods (seed 3) — never trained on, the certification set; `HOLD2` 16 periods (seeds 8000–8015). Within-seen validation uses positions [384, 512) on training blocks. Certification is scored on positions [64, 512).

Fit (all three students; identical recipe). Adam, lr = 1×10^{-3}, 4000 steps, fp32, TF32 off, pure MSE loss on the projected-complement residual. **No ridge / weight decay / other explicit regularization** — the linear student's only "regularizer" is its capacity (it is too small to memorize). Per-student seeds RUNG_SEED {L0:11, L1:22, L2:33} + master 20260706. Downsampled loss curves (40 points/student) are frozen (Fig. 4).

Architectures / parameters. Linear `nn.Linear(1537, 768)` — **1,181,184** params. MLP `[1537→768, GELU, 768→768]` — 1,771,776. Attention `inp 1537→256, 2 blocks × 4 heads, out 256→768` — 1,776,384.

Shuffled-target twin. Same features, targets permuted across the rep-era training tokens; the pass rule is `real KL ≤ 0.5 × twin KL`.

Results (verified).

student	SACRED KL	floor (rep_b5 recal)	SACRED-R^2	within-seen KL	train-R^2	twin KL	copy fidelity	certified
Linear	**0.11172**	0.1279	0.7716	0.01296	0.8758	1.53558 (13.75×)	0.9636	**YES**
MLP	0.35442	0.1279	0.24	0.00095	0.9782	8.56376	0.8298	no (memorizes)
Attention	0.34468	0.1279	0.3285	0.00127	0.9667	5.12161	0.8437	no (memorizes)

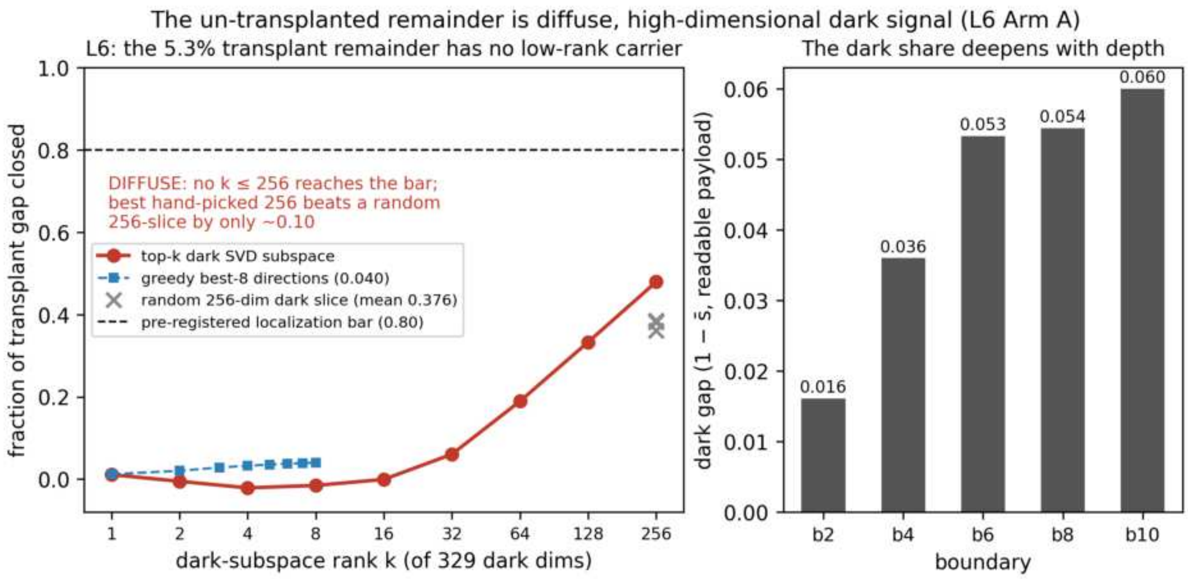
The high-capacity students fit training content nearly perfectly (train-R^2 0.98/0.97, within-seen KL ≈ 0.001) yet fail on new periods (SACRED KL ≈ 0.35) — the capacity-hurts falsifier. Onset rungs: b6 SACRED 0.02644 ≤ floor 0.08057 (copy fid 0.9995); b7 0.06665 ≤ 0.10724 (0.9711). Thin-margin discharge (V7): frozen linear student over 5 fresh batches (80 never-seen blocks, seeds 9000–9400) — median 0.11814, 4/5 under floor, worst 1.05×. Drop-m0 ablation: retraining on {x2, ecur} alone gives 0.11105 (slightly *better*) — the layer-2 state + current token are the load-bearing inputs; m0 is not needed.

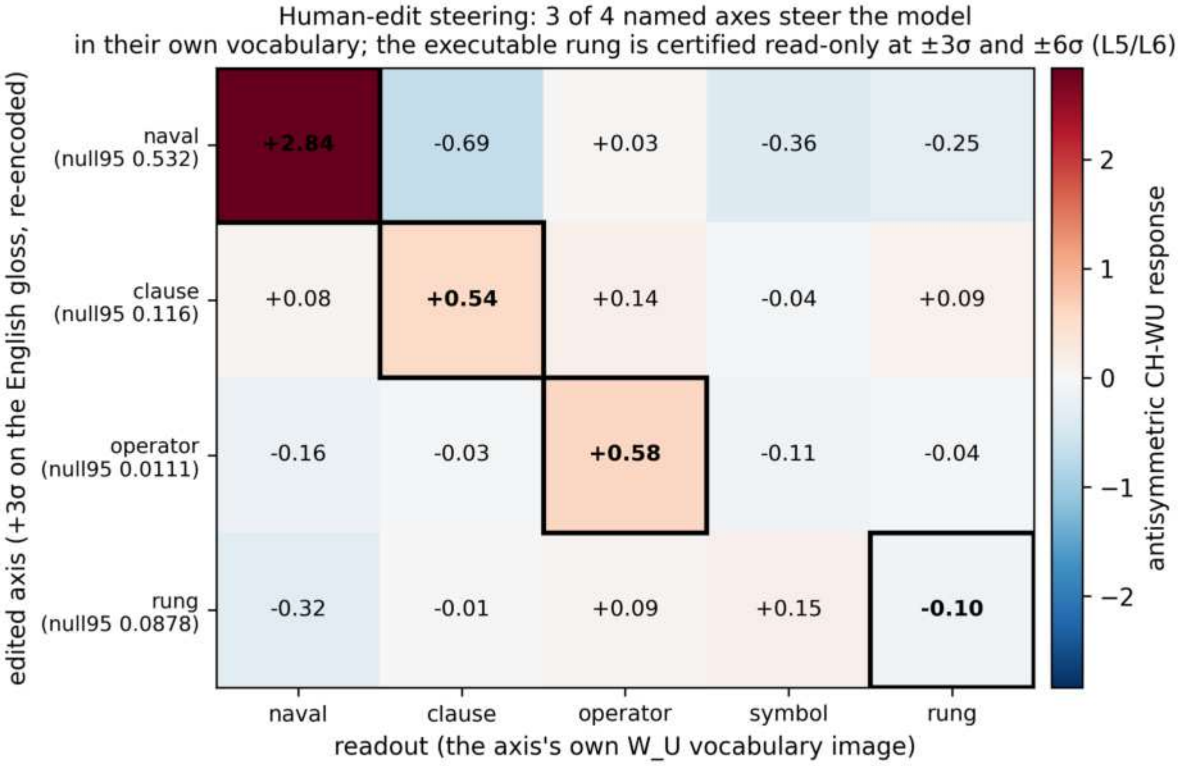
Not recoverable from disk (stated honestly): per-step gradient statistics; per-step wall-clock; the exact twin-permutation index trajectory (reproducible only by re-running with the frozen seeds); per-layer activation statistics during training.

Figures

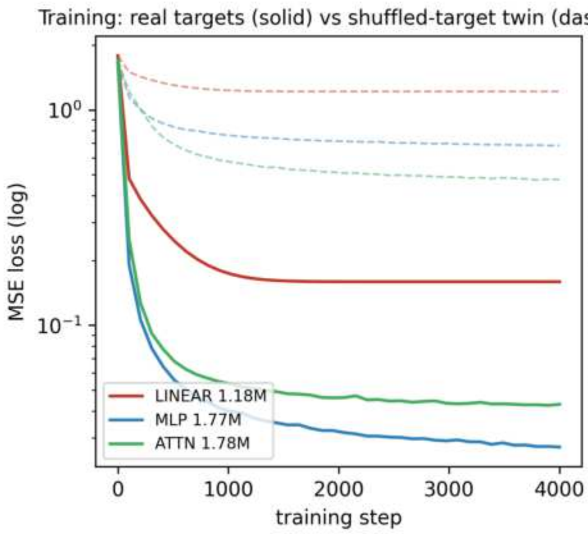
- **Fig. 1** paper_figs/fig1_nat_collapse — unexplained mass and gap cells per round, both meters (11.18 → 0.000 recal / 0.283 legacy).
- **Fig. 2** paper_figs/fig2_verdict_heatmaps — final 39-cell KL/floor heatmaps, recal + legacy.
- **Fig. 3** paper_figs/fig3_kl_vs_rank — the late tail sells out gradually (V3); rep_b12 plateaus above the legacy floor at the allowance cap (V4).
- **Fig. 4** paper_figs/fig4_surrogate_falsifier — surrogate training vs shuffled-target twins; the capacity-hurts falsifier at BUS[5].
- **Fig. 5** paper_figs/fig5_speak_confusion — the human-edit confusion matrix (T3); plotted values unchanged (regenerated 2026-07-06 from the frozen _l4_result.json). The in-figure title annotation carries the pre-audit shorthand ("certified read-only at ±3σ and ±6σ") — the figure is a frozen byte-identical artifact; the precise two-null verdict (steering-unusable: not separable from the honest N = 20 floor at ±6σ, marginally above it at ±3σ, sub-linear scaling) is §6.6–6.7.

- **Fig. 6** paper_figs/fig6_dark_rank_ladder — the 5.3% transplant remainder is DIFFUSE (L6): gap closure vs dark-SVD rank (no k ≤ 256 reaches the pre-registered 0.80 bar; greedy-8 and the random-256-slice control shown) and the per-boundary dark gap (deepens with depth). Generated from the frozen _l6_result.json.

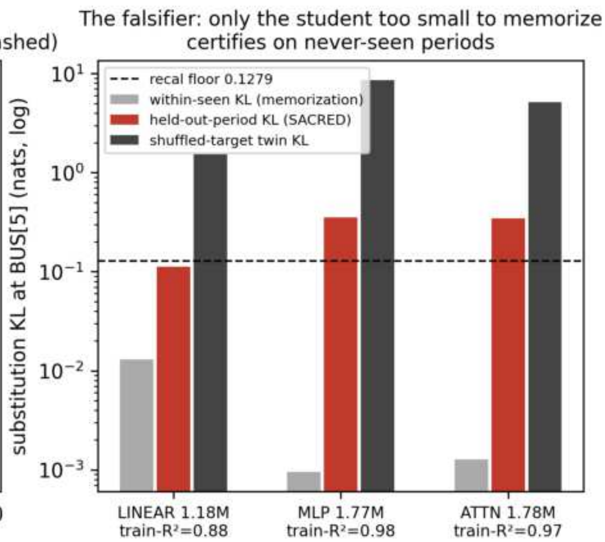




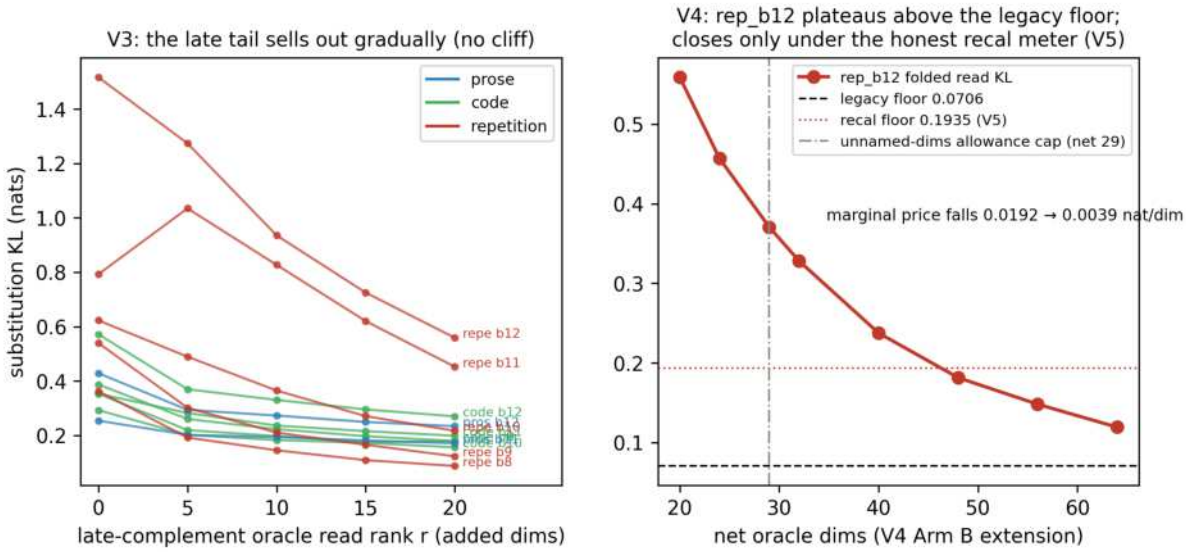
Human-edit steering: 3 of 4 named axes steer the model
in their own vocabulary; the executable rung is certified read-only at ±3σ and ±6σ (L5/L6)
edited axis (+3σ on the English gloss, re-encoded)
naval (null95 0.532)
clause (null95 0.116)
operator (null95 0.0111)
rung (null95 0.0878)
+2.84
-0.69
+0.03
-0.36
-0.25
+0.08
+0.54
+0.14
-0.04
+0.09
-0.16
-0.03
+0.58
-0.11
-0.04
-0.32
-0.01
+0.09
+0.15
-0.10
naval
clause
operator
symbol
rung
readout (the axis's own W_U vocabulary image)
antisymmetric CH-WU response

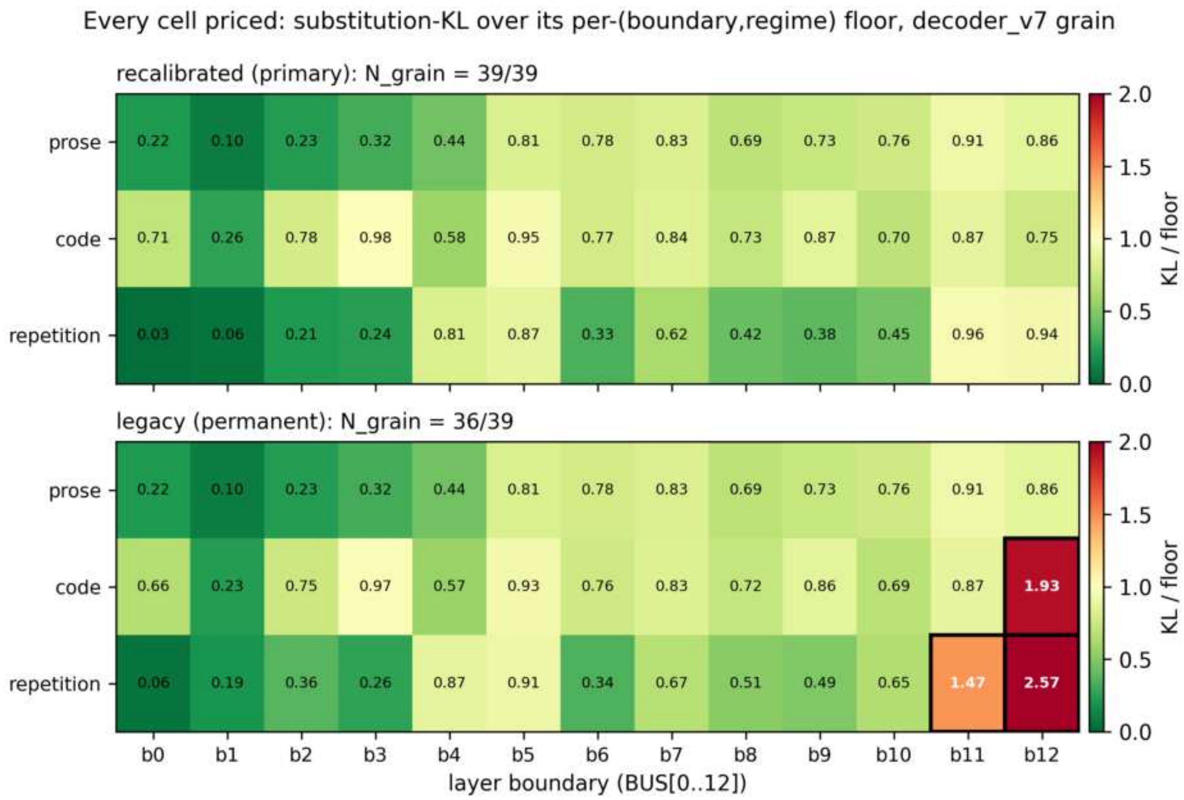


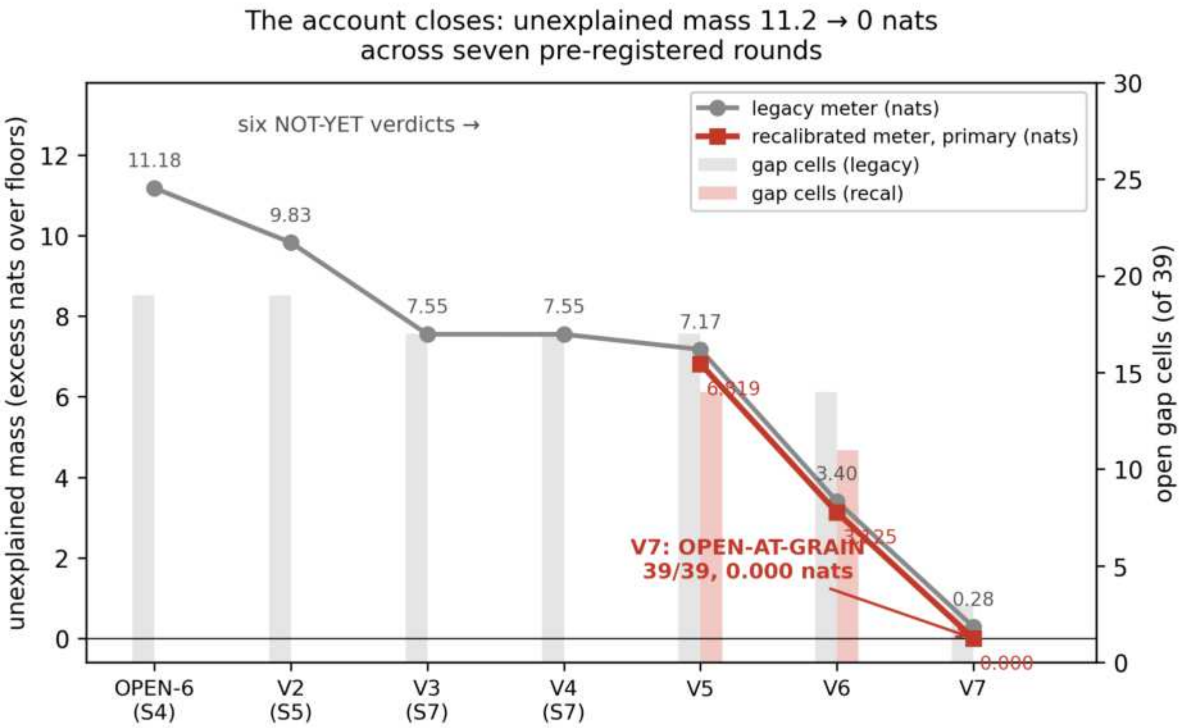
Training: real targets (solid) vs shuffled-target twin (dashed)
MSE loss (log)
training step
LINEAR 1.18M
MLP 1.77M
ATTN 1.78M



The falsifier: only the student too small to memorize
certifies on never-seen periods
substitution KL at BUS[5] (nats, log)
recal floor 0.1279
within-seen KL (memorization)
held-out-period KL (SACRED)
shuffled-target twin KL
LINEAR 1.18M
train-R²=0.88
MLP 1.77M
train-R²=0.98
ATTN 1.78M
train-R²=0.97


V3: the late tail sells out gradually (no cliff)
substitution KL (nats)
late-complement oracle read rank r (added dims)
prose
code
repetition
repe b12
repe b11
code b12
repe b9
repe b8
V4: rep_b12 plateaus above the legacy floor;
closes only under the honest recal meter (V5)
rep_b12 folded read KL
legacy floor 0.0706
recal floor 0.1935 (V5)
unnamed-dims allowance cap (net 29)
marginal price falls 0.0192 → 0.0039 nat/dim
net oracle dims (V4 Arm B extension)


Every cell priced: substitution-KL over its per-(boundary,regime) floor, decoder_v7 grain
recalibrated (primary): N_grain = 39/39
prose 0.22 0.10 0.23 0.32 0.44 0.81 0.78 0.83 0.69 0.73 0.76 0.91 0.86
code 0.71 0.26 0.78 0.98 0.58 0.95 0.77 0.84 0.73 0.87 0.70 0.87 0.75
repetition 0.03 0.06 0.21 0.24 0.81 0.87 0.33 0.62 0.42 0.38 0.45 0.96 0.94
legacy (permanent): N_grain = 36/39
prose 0.22 0.10 0.23 0.32 0.44 0.81 0.78 0.83 0.69 0.73 0.76 0.91 0.86
code 0.66 0.23 0.75 0.97 0.57 0.93 0.76 0.83 0.72 0.86 0.69 0.87 1.93
repetition 0.06 0.19 0.36 0.26 0.87 0.91 0.34 0.67 0.51 0.49 0.65 1.47 2.57
KL / floor
b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 b10 b11 b12
layer boundary (BUS[0..12])



The account closes: unexplained mass 11.2 → 0 nats
across seven pre-registered rounds
six NOT-YET verdicts →
legacy meter (nats)
recalibrated meter, primary (nats)
gap cells (legacy)
gap cells (recal)
unexplained mass (excess nats over floors)
open gap cells (of 39)
11.18
9.83
7.55
7.55
7.17
6.819
3.40
3.125
0.28
0.000
V7: OPEN-AT-GRAIN
39/39, 0.000 nats
OPEN-6 (S4)
V2 (S5)
V3 (S7)
V4 (S7)
V5
V6
V7